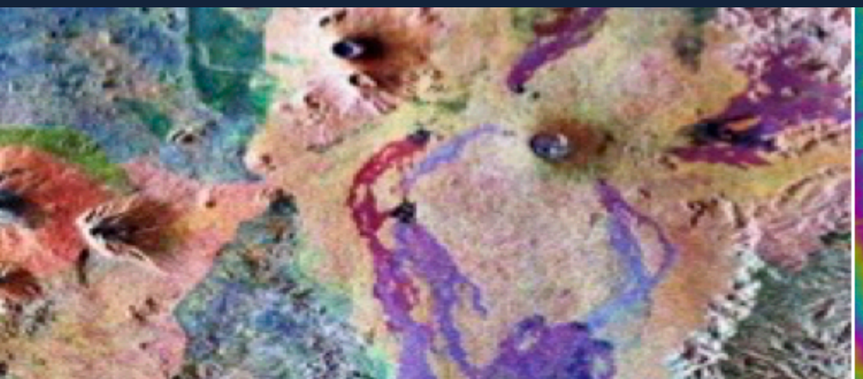
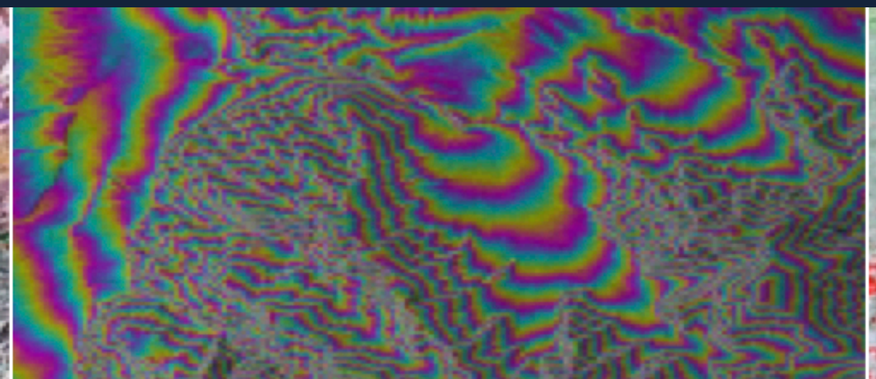
SAR PRINCIPLES AND PHENOMENOLOGY

Synthetic Aperture Radar Interferometry

AKSHAY RAJ MANOCHA





THIS LECTURE

What you will learn

01 What is radar?

02 Basic definitions

03 What is synthetic aperture radar (SAR)?

04 Properties of radar images

05 What is SAR interferometry (InSAR)?

06 What other SAR modalities are important?

07 What is range-Doppler geometry, why is it important, and how does it relate to geographic coordinates?

LECTURE OUTLINE

First segment: radar imaging fundamentals

1

Basic principles of radar and SAR

Waves, range and ground resolution, beams, Doppler, aperture synthesis and SAR modes

2

Properties of radar images

What radar measures, scattering, imaging geometry, wavelength, polarization, the radar equation and speckle

INTRODUCTION AND MOTIVATION

How can a radar that lights up a huge footprint on the ground deliver very fine-resolution images, and measure ground motion over time to mm–cm accuracy?

SAR and InSAR are technically complex. Even experts find them a source of wonder. Let’s start by looking at an “image” of the raw data, before any processing.

Motivation – Understanding the processes and phenomenology of radar interferometry

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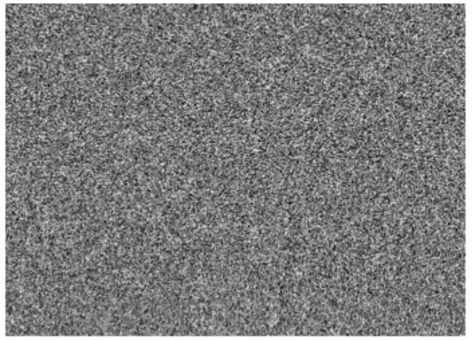
Satellite Observation



First Time

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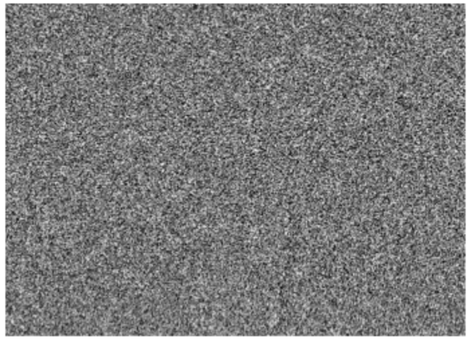

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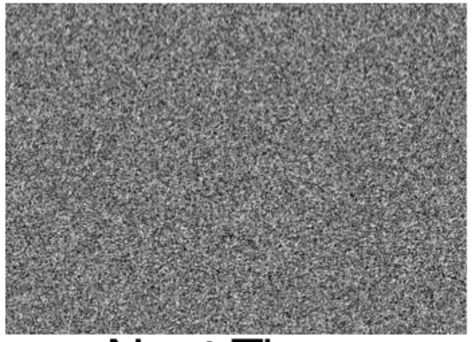
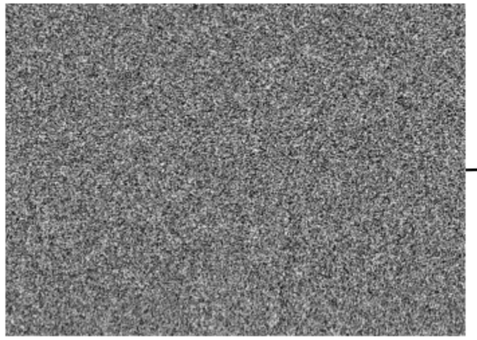


Next Time

Motivation – Understanding the processes and phenomenology of radar interferometry

Satellite Observation

Magic of imaging



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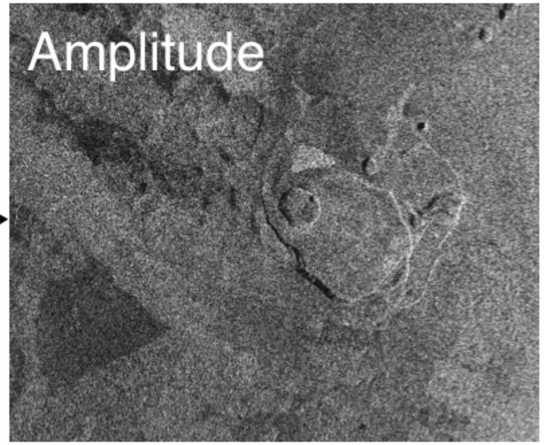

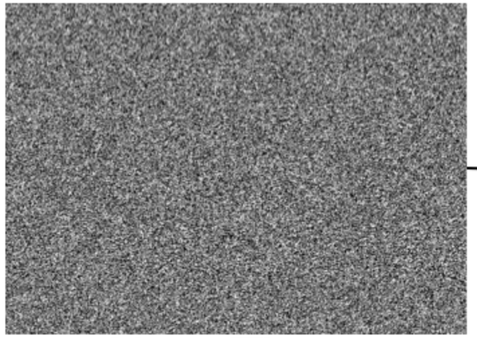

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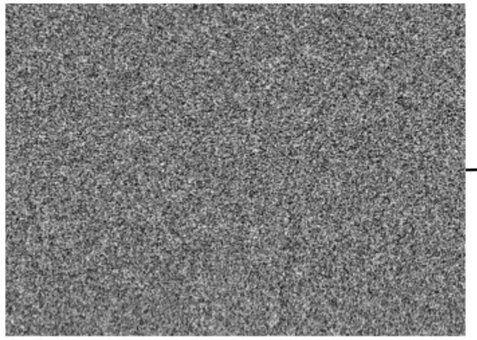

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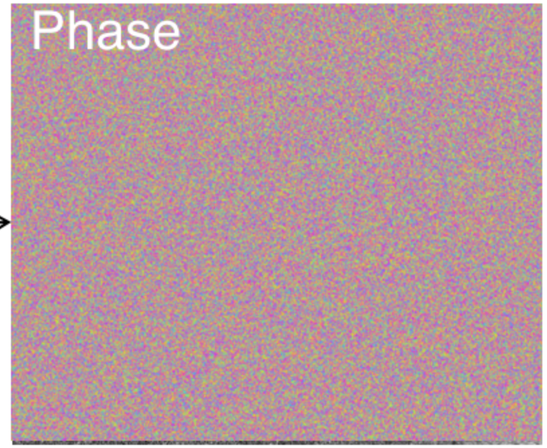

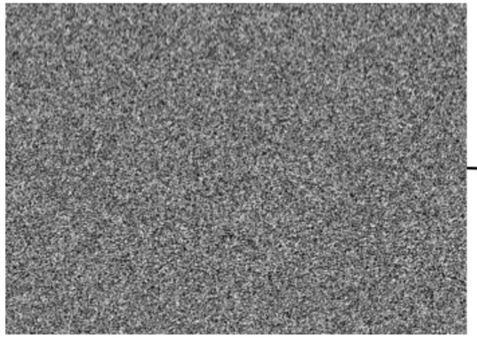

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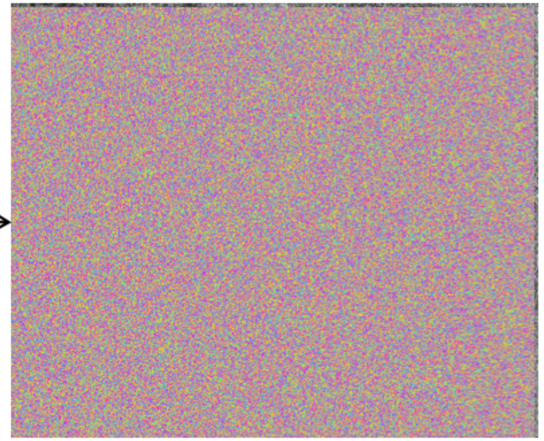

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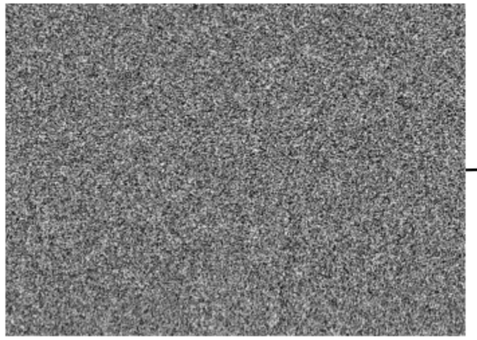

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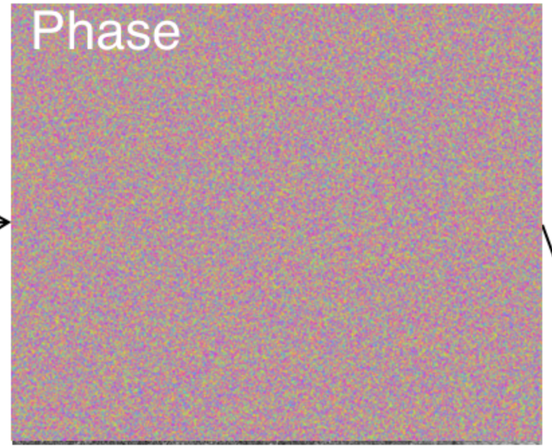

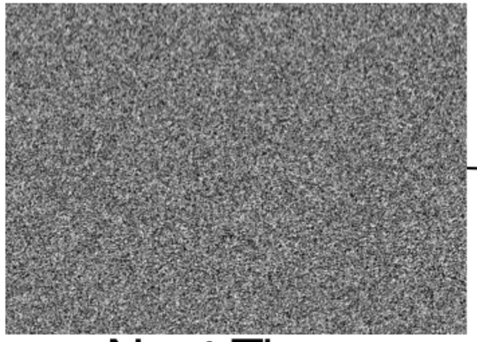

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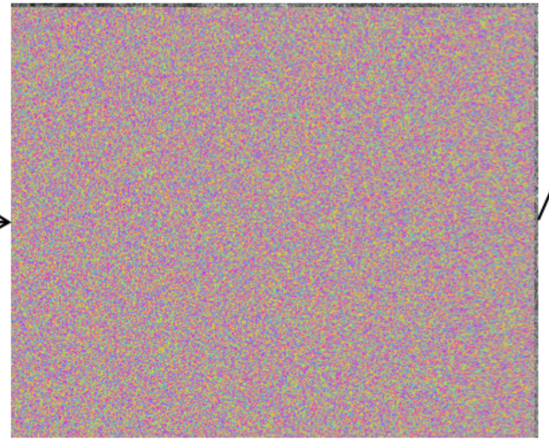

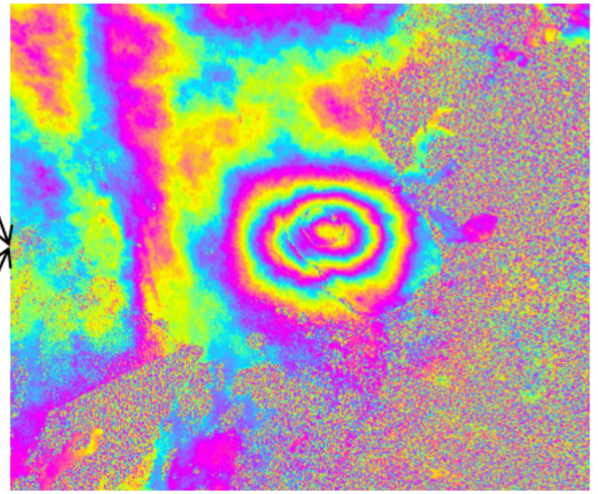

Magic of interferometry

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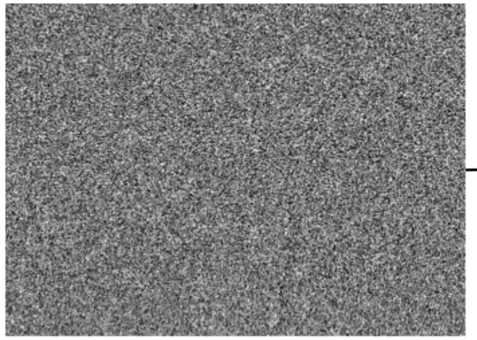

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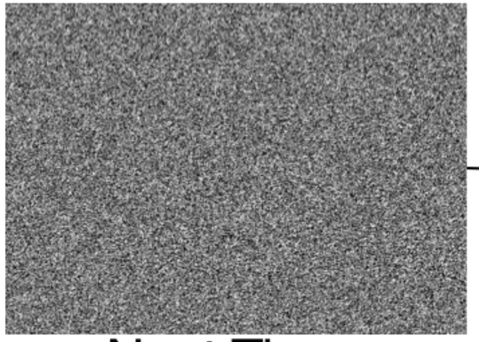
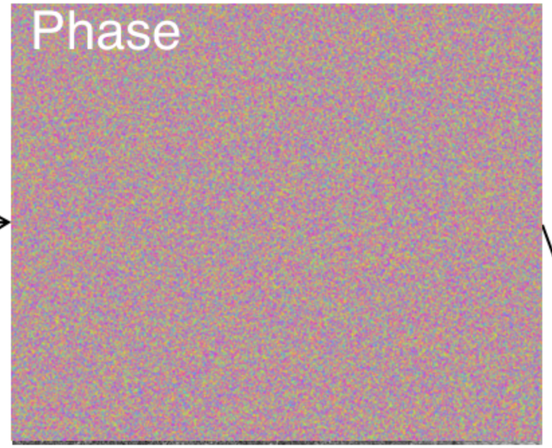
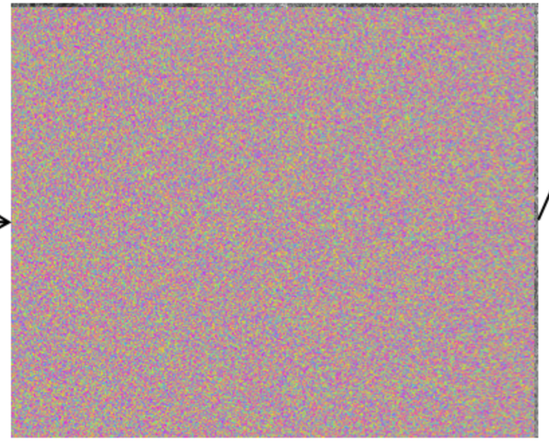
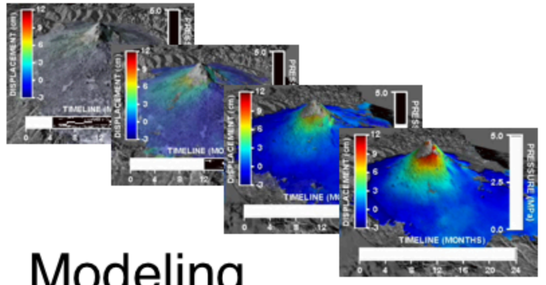


Next Time

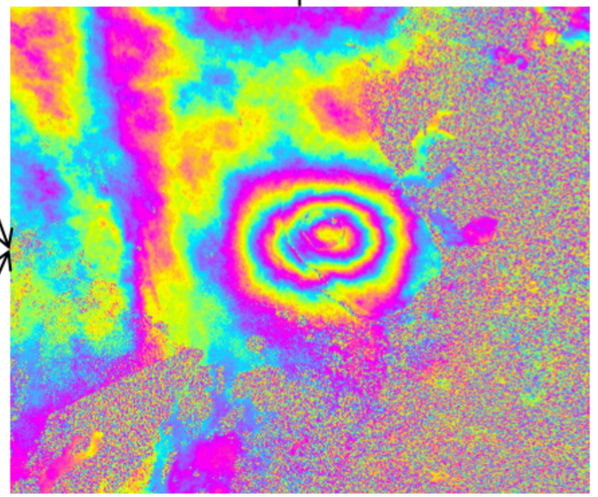
Magic of imaging







Modeling



Magic of interferometry

01

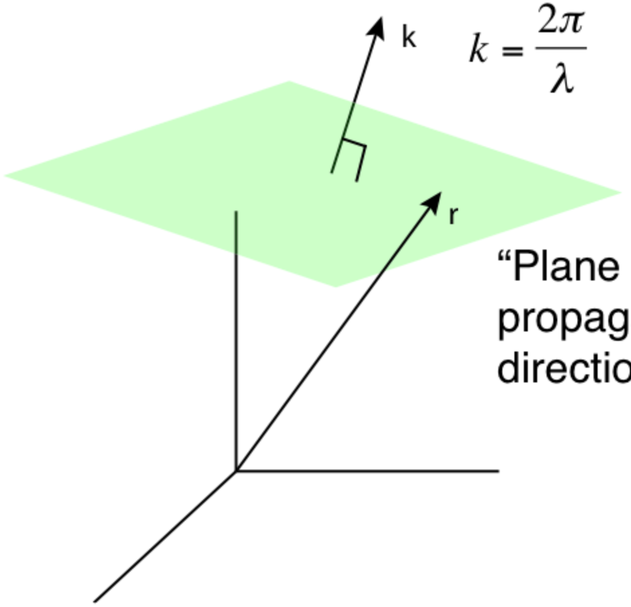
PART 1 · RADAR IMAGING FUNDAMENTALS

Basic principles of radar and SAR

Waves and radar · range and ground resolution · beams and Doppler · aperture synthesis · SAR modes

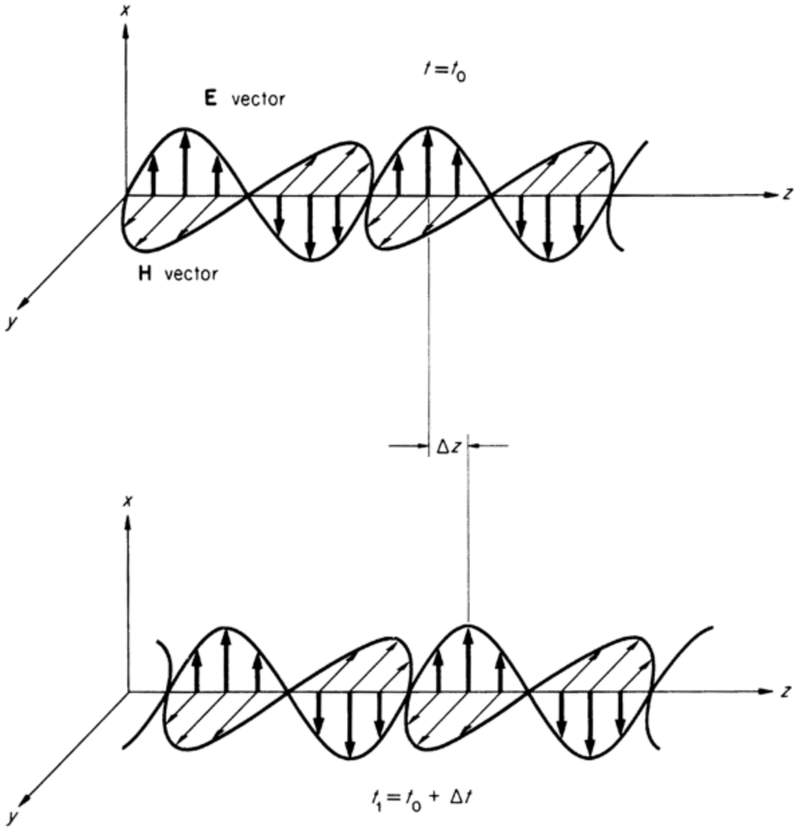
Wave Properties of Light

- Radar waves, like lightwaves, are electromagnetic energy that propagates



$$k = \frac{2\pi}{\lambda}$$

"Plane wave" propagating in direction k

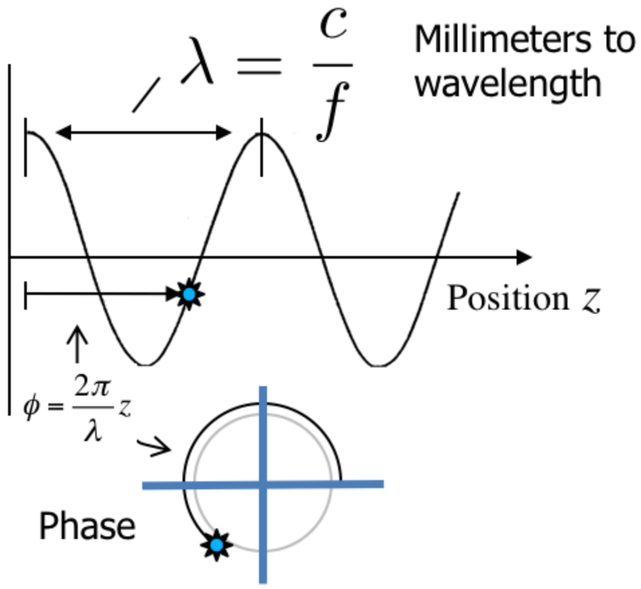


Energy propagates

"Phase" ↓ frequency ↓

$$E(x,y,z,t) = E_0 e^{j\phi(x,y,z,t)} = E_0 e^{j(\omega t - \vec{k}\cdot\vec{r})}$$

Radar and Light Waves

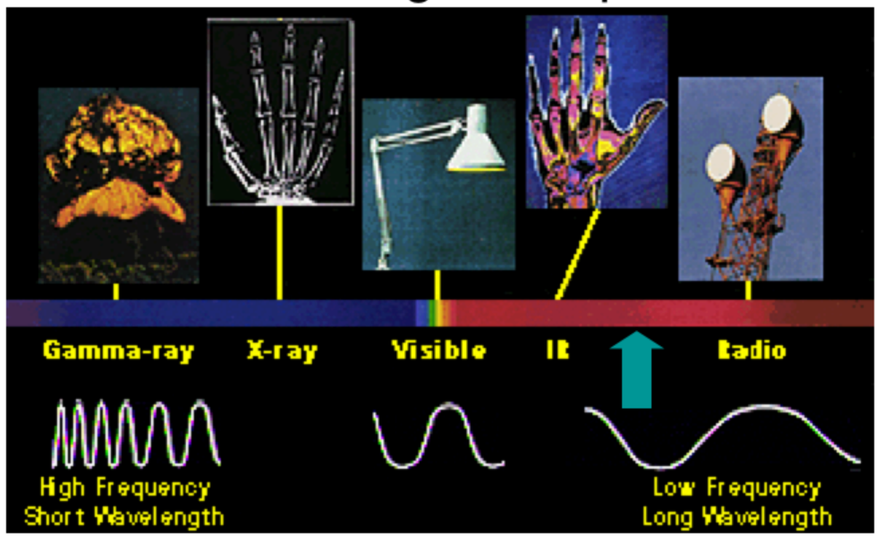


- Radars operate at microwave frequencies, an invisible part of the electromagnetic spectrum
- Microwaves have wavelengths in the millimeter to meter range
- Like lasers, radars are coherent; we can determine the signal's phase

Common Radar Frequency Bands

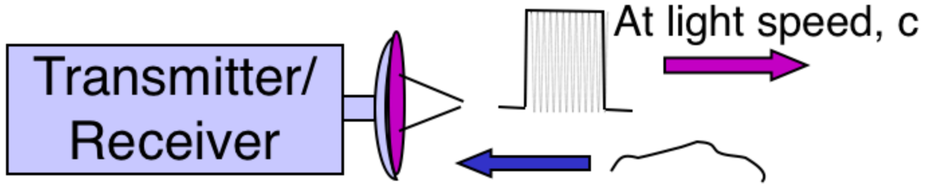
Band	Ka	Ku	X	C	S	L	P
Wavelength (cm)	1	2	3	6	12	24	75
Frequency (G-cycles/s)	30	15	10	5	2.5	1.2	0.4

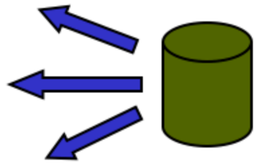
The Electromagnetic Spectrum



tiny 100's μm mm's to m's

The Radar Concept





Object scatters energy back to radar

- Much like sound waves, radar waves carry information that echoes from distant objects
- The time delay of the echo measures the distance to the object
- The changes of the message in the echo determines the object characteristics

WORKED EXAMPLES

Parameters used in the examples

Geometry and platform

Parameter	Symbol	Value
Wavelength (L-band)	λ	0.24 m
Antenna length	L_a	10 m
Antenna width	W_a	2 m
Off-nadir boresight angle	θ_l	30°
Azimuth squint angle	θ_sq	0°
Spacecraft velocity	v_sc	7,500 m/s
Spacecraft altitude	h_sc	750 km

Signal and power

Parameter	Symbol	Value
Range bandwidth	B_r	20 MHz
Pulse duration	τ_r	20 µs
Pulse rate	f_p	1,600 Hz
Peak transmit power	P_T	4,000 W
Noise temperature	T_r	300 K
Corner reflector size	L_cr	2.4 m

Speed of light, c

$299{,}792{,}458\ \mathrm{m/s}$

Boltzmann constant, k

$1.3806 \times 10^{-23}\ \mathrm{J/K}$ (−228.6 dB)

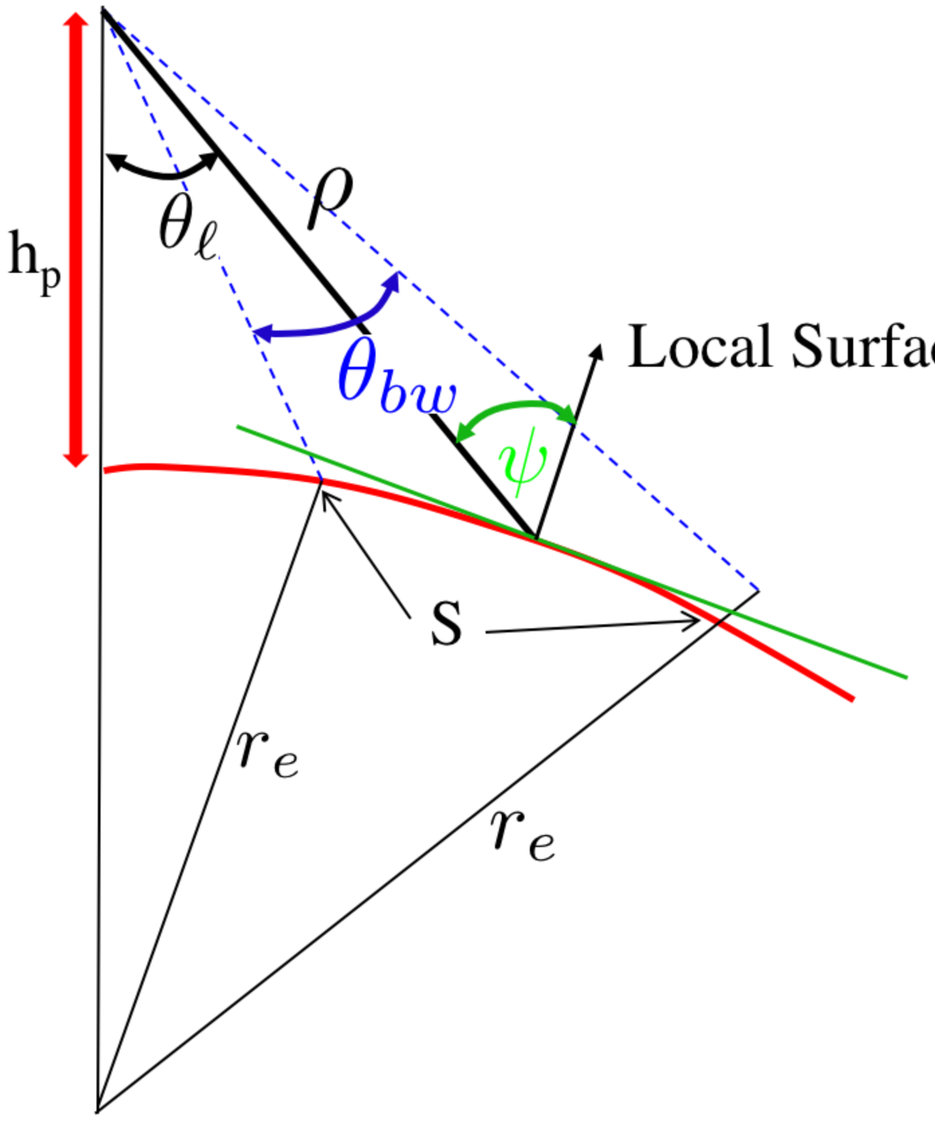
Gravitational constant, G

$6.672 \times 10^{-11}\ \mathrm{m^3\ kg^{-1}\ s^{-2}}$

Earth’s mass, M_E

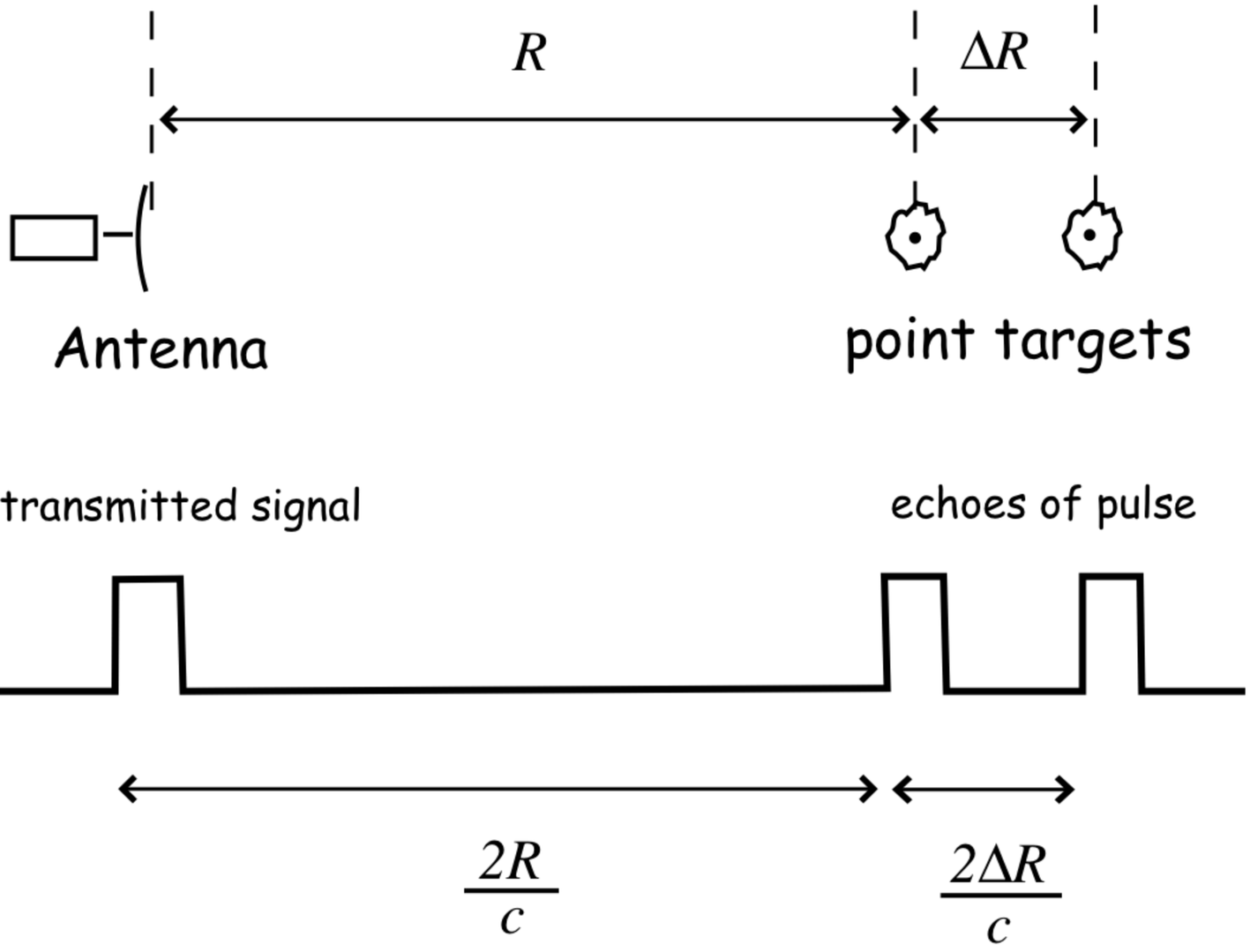
$5.9742 \times 10^{24}\ \mathrm{kg}$

Radar Geometric Terminology



θ = Look Angle
ψ = Incidence Angle
ρ = Range
θ_{bw} = beamwidth
S = Swath width
r_e = Radius of Earth
h_p = Height of platform

Achieving resolution in range



If possible, use a narrow pulse to resolve close targets

RANGE CODING

Coding the pulse buys resolution

If a narrow enough pulse can’t be sent with enough power, the radar transmits a **coded waveform** that spreads the pulse over a wider bandwidth. Resolution is then the reciprocal of the bandwidth.

Uncoded pulse of duration τ_r

$$\Delta\rho = c \cdot \tau_r / 2$$

Pulse coded to bandwidth B_r

$$\Delta\rho = c / (2 \cdot B_r)$$

1,499 m

Uncoded pulse, $\tau_r = 10\ \mu s$

7.5 m

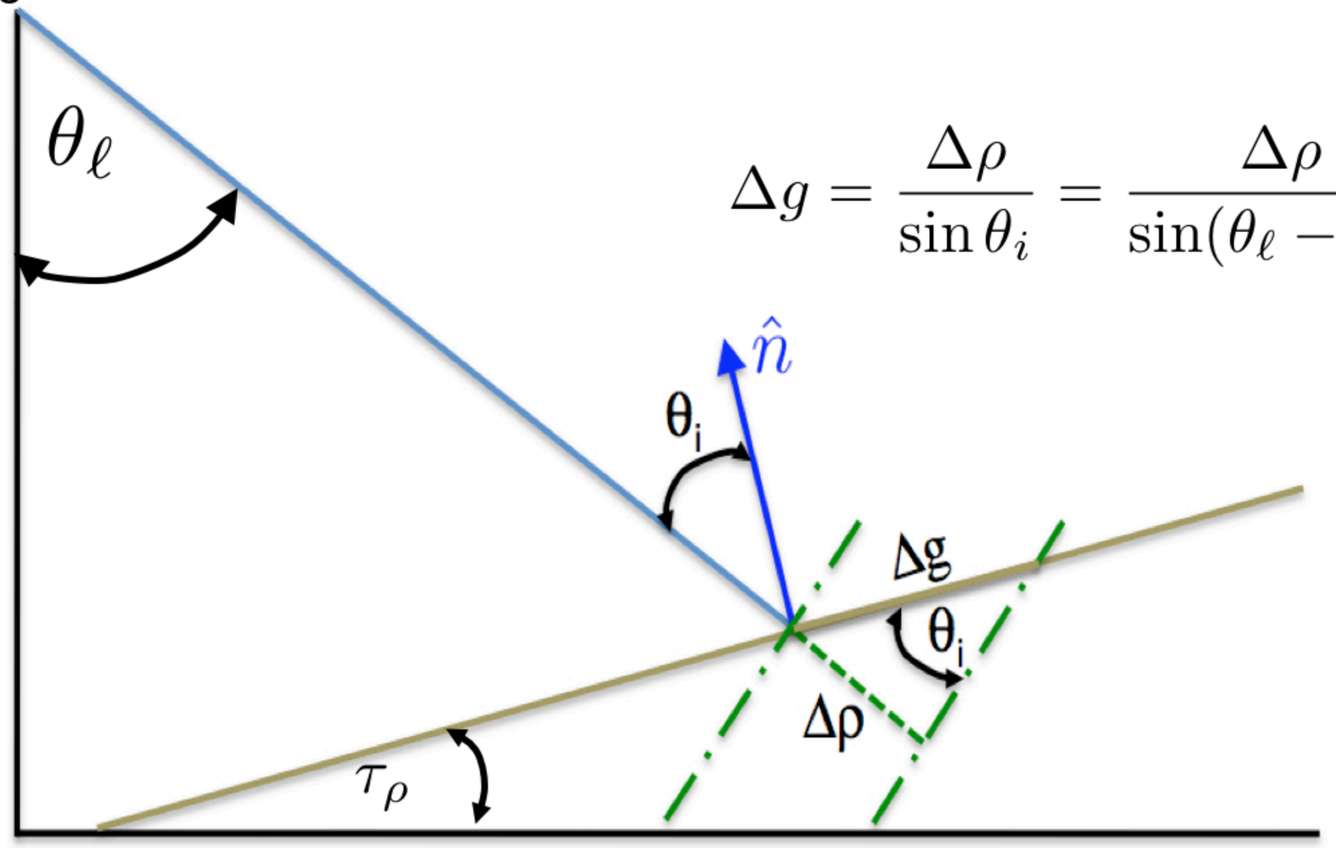
Coded pulse, $B_r = 20$ MHz

Altitude $h_{sc} = 750$ km and look angle $\theta_l = 30°$ give a slant range to the beam centre of $\rho = h_{sc} / \cos\theta_l \approx 866$ km.

Ground Resolution

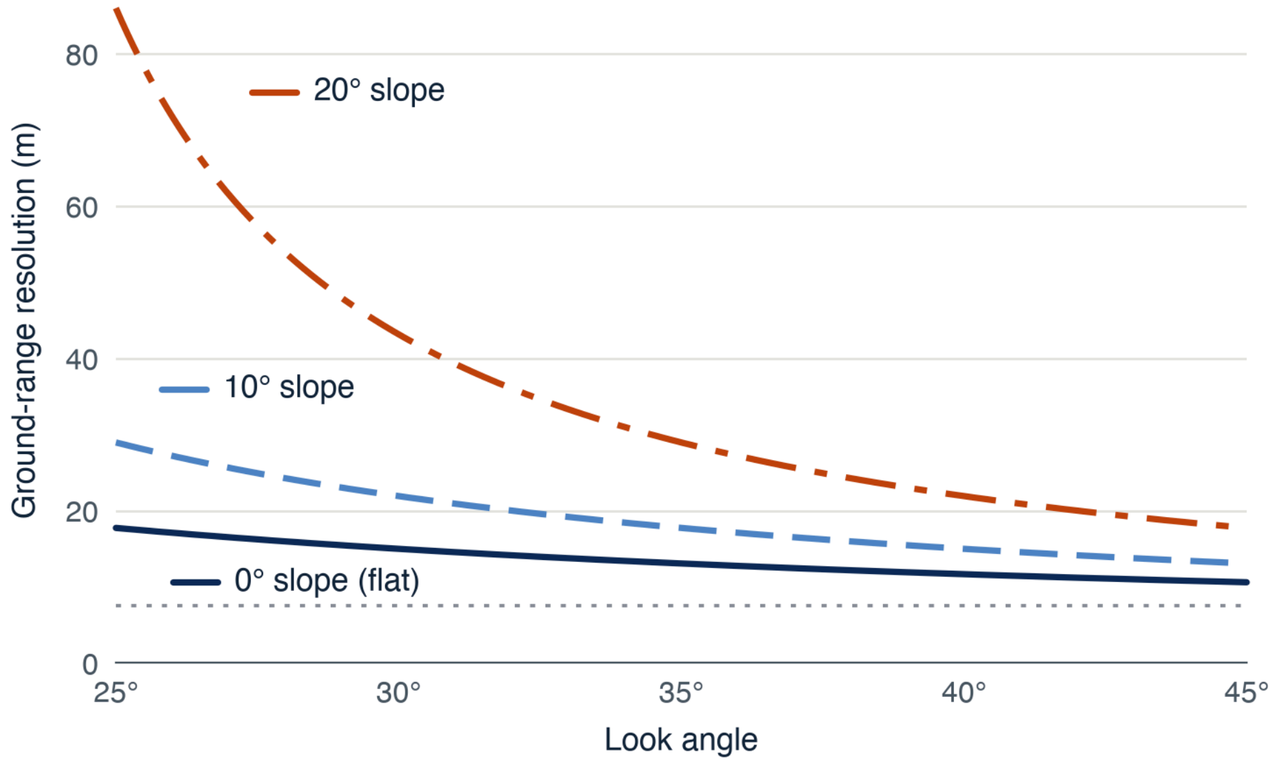
- The ground resolution, Δg, depends on the range resolution and incidence angle, θ_i.
- Incidence angle is the angle between the wave propagation direction or line-of-sight and the normal to the surface.

$$\Delta g = \frac{\Delta\rho}{\sin\theta_i} = \frac{\Delta\rho}{\sin(\theta_\ell - \tau_\rho)}$$



TERRAIN EFFECTS

Ground resolution, look angle and slope



Ground-range resolution

$$\Delta g = \Delta \rho \,/\, \sin(\theta - \tau_p)$$

θ = look angle, τ_p = slope facing the radar

AT 25° LOOK ANGLE

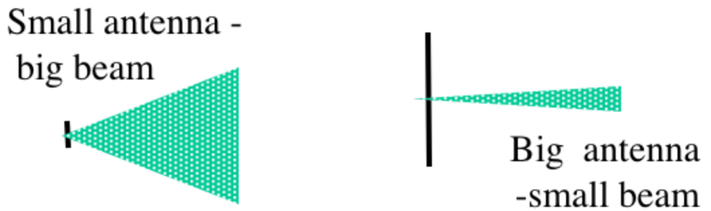
17.7 m flat ground

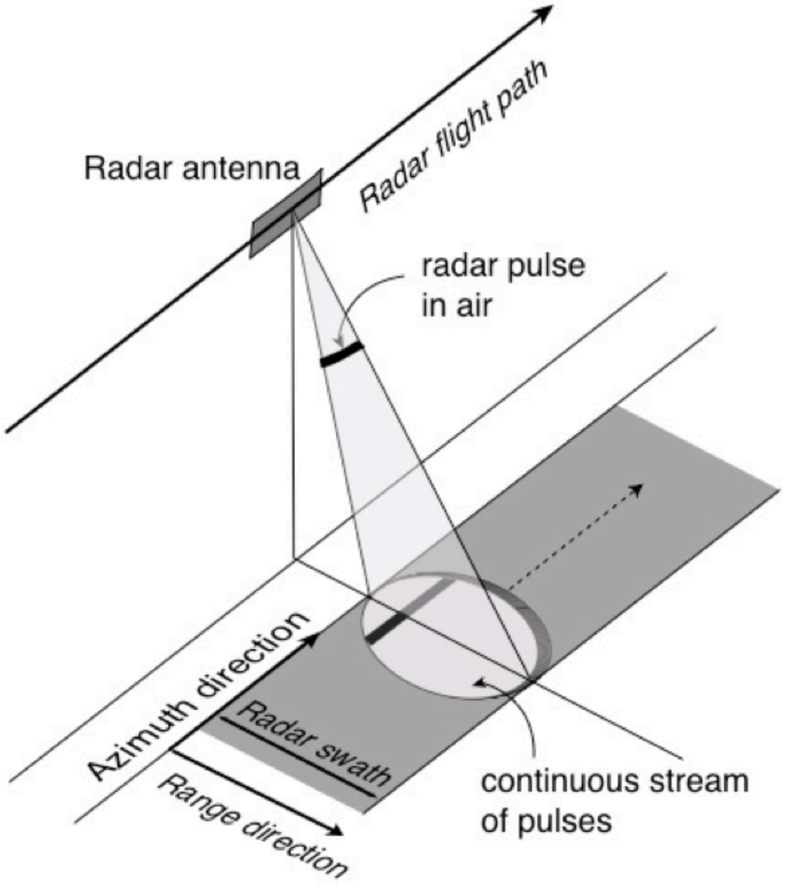
29.0 m 10° slope

86.0 m 20° slope

Dotted line: slant-range resolution, 7.5 m (B_r = 20 MHz)

Radar on a Moving Platform





- Pulses are transmitted from the radar platform as it moves along its flight path
- Each pulse has finite extent in time, illuminating a narrow strip of ground as it sweeps through the antenna beam
- Some of the energy from the ground is scattered back to the radar instrument

Antenna Beamwidth

$$\theta_{3dB} = \frac{\lambda}{L}$$

Footprint Size on Ground

$$\Delta X_{az} \approx \rho\theta_{3dB_{az}} = \frac{\rho\lambda}{L_{az}}$$

$$\Delta X_{\rho} \approx \rho\theta_{3dB_{\rho}} = \frac{\rho\lambda}{L_{\rho}}$$

WORKED EXAMPLE

Example beam dimensions

$$\theta_{3dB} \approx 0.866\ \lambda / L_{footprint} \approx \rho \cdot \theta_{3dB}$$

1.19°

Azimuth beamwidth (antenna length L_a = 10 m)

5.95°

Elevation beamwidth (antenna width W_a = 2 m)

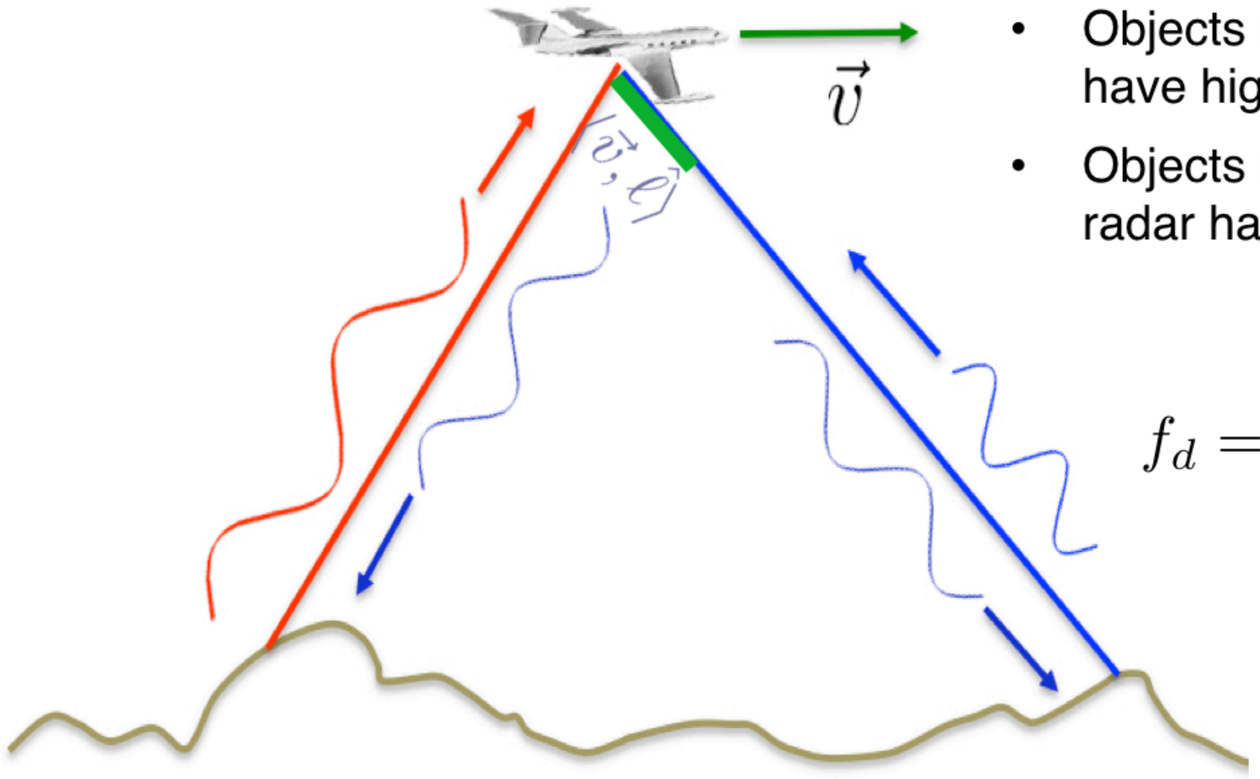
18.0 km

Azimuth beam extent at ρ = 866 km

103.9 km

Elevation beam extent, projected onto the ground (÷ cos θ_l)

Doppler Shift

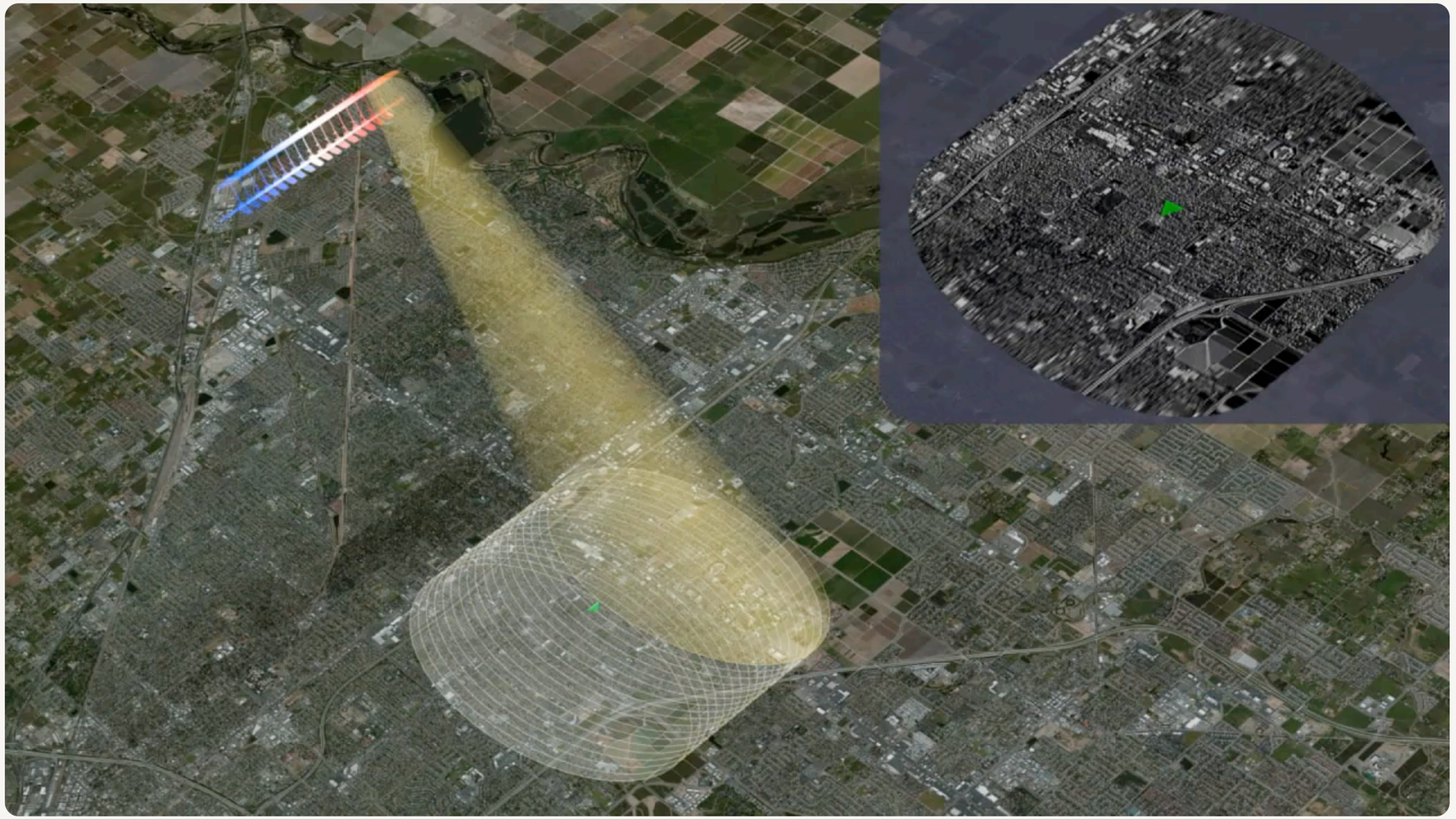


- Objects moving relative to a radar experience a frequency shift called the Doppler shift.
- Objects moving toward the radar have higher frequencies
- Objects moving away from the radar have lower frequencies.

$$f_d = \frac{2\langle \vec{v}, \hat{\ell} \rangle}{\lambda} = \frac{2v \cos\theta_{sq}}{\lambda}$$

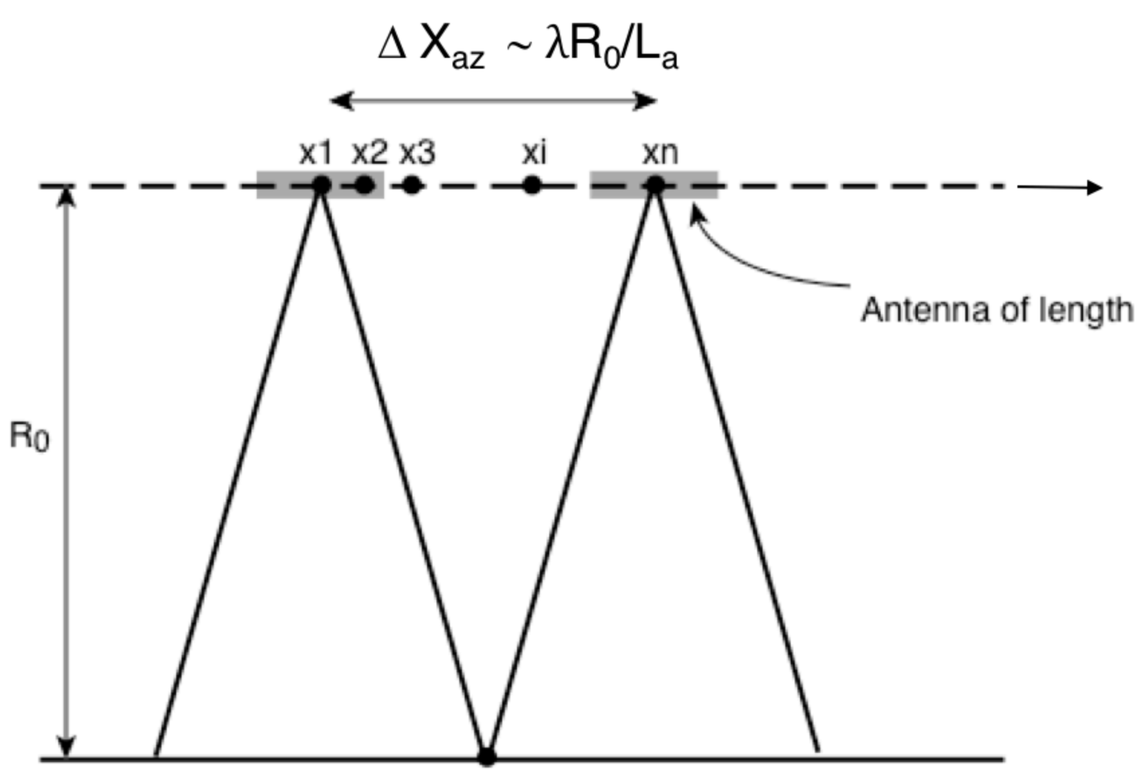
ANIMATION

SAR imaging in action



As the satellite moves, its beam sweeps the ground pulse after pulse. The inset shows the focused radar image forming.

Azimuth Resolution from Aperture Synthesis



- The illuminated extent of the ground is $\Delta X_{az} \sim \lambda R_0/L_a$
- The synthetic aperture is $2^*\Delta X_{az}$ lengthens as R_0 increases…
- …which decreases the azimuth synthetic aperture's angular beamwidth $\lambda/(2^*\Delta X_{az})$ in proportion…
- …but the spatial resolution on the ground
  - $(\lambda R_0/ (2^*\Delta X_{az}) = L_a/2$
  - $L_a/2$ is constant

WORKED EXAMPLE

Azimuth resolution from a synthetic aperture

Two-way propagation doubles the Doppler bandwidth across the azimuth beam, so the synthetic aperture is effectively twice the illuminated extent.

10 m
Real antenna length, L_a

→

18.0 km
Beam extent on the ground

→

36.0 km
Synthetic aperture (2 × beam extent)

→

5.0 m
Azimuth resolution ≈ L_a / 2

Effective synthetic beamwidth: 0.331 millidegrees. The resolution does not depend on range.

TEST YOUR UNDERSTANDING

Radar resolution

What is the intrinsic resolution of a stripmap SAR image with **40 MHz** range bandwidth and an antenna **2 m** in elevation and **12 m** in the flight direction?

A 12 m in range and 12 m in flight direction

B 1 m in range and 6 m in flight direction

C 3 m in range and 1 m in flight direction

D 3 m in range and 6 m in flight direction

ANSWER

D: 3 m in range, 6 m along track

Range: set by bandwidth

$$\Delta\rho = c / (2 \cdot B_r)$$

$= 3 \times 10^8 / (2 \times 40 \times 10^6) \approx 3.75$ m. The closest option is 3 m.

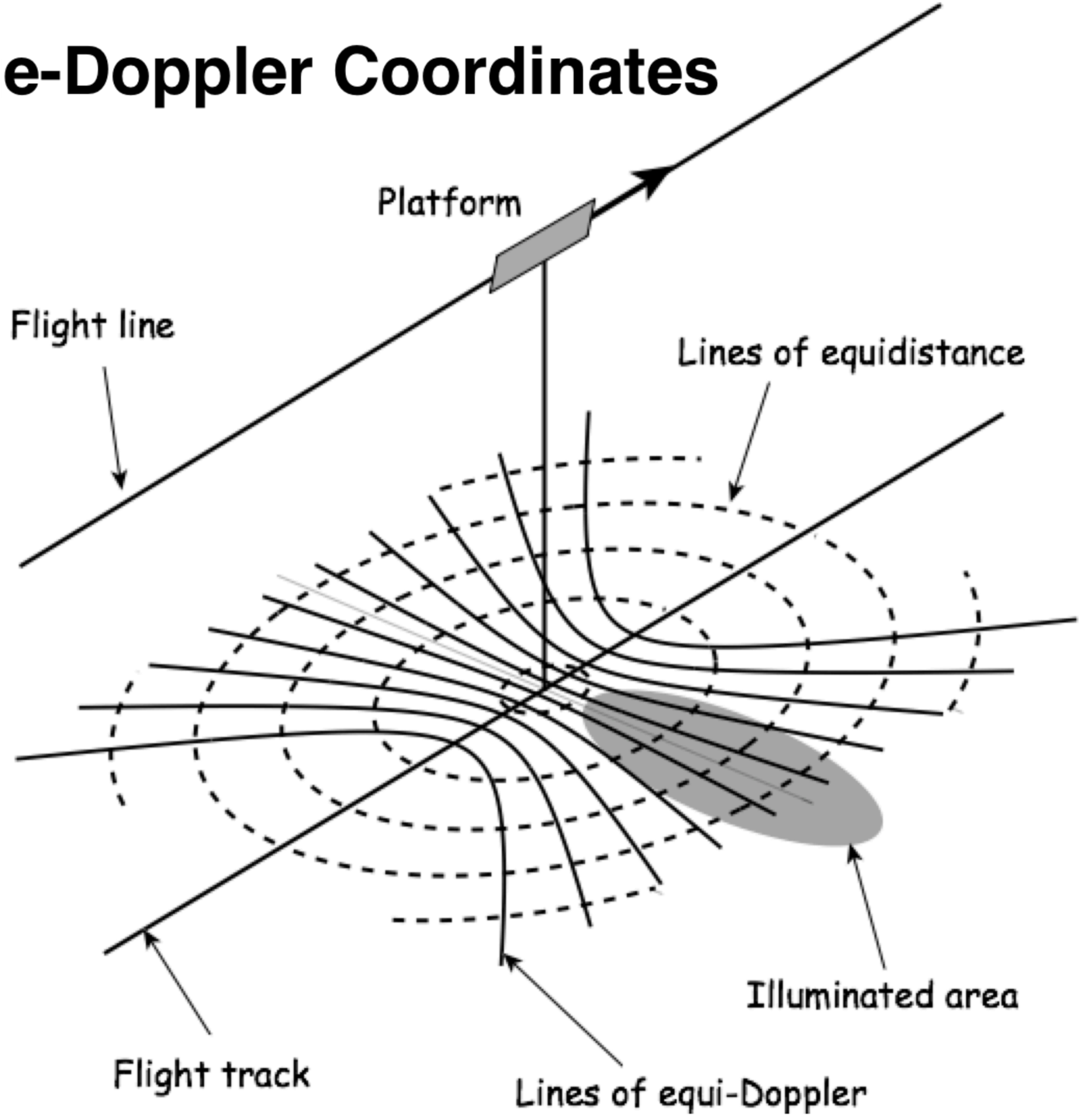
Flight direction: set by antenna length

$$\delta x \approx L_a / 2$$

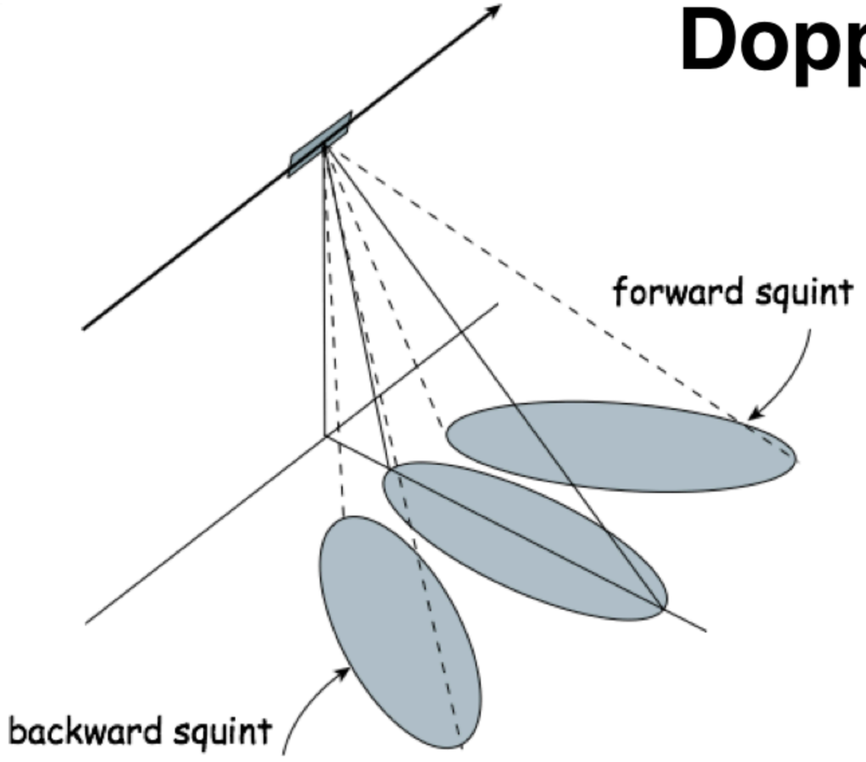
$= 12$ m $/ 2 = 6$ m.

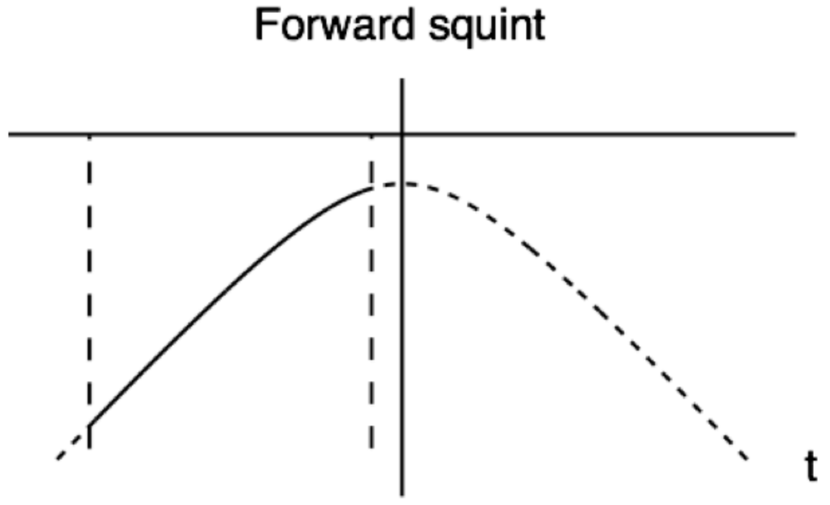
The 2 m elevation dimension sets the beam width across track, and so the swath. It does not set the resolution.

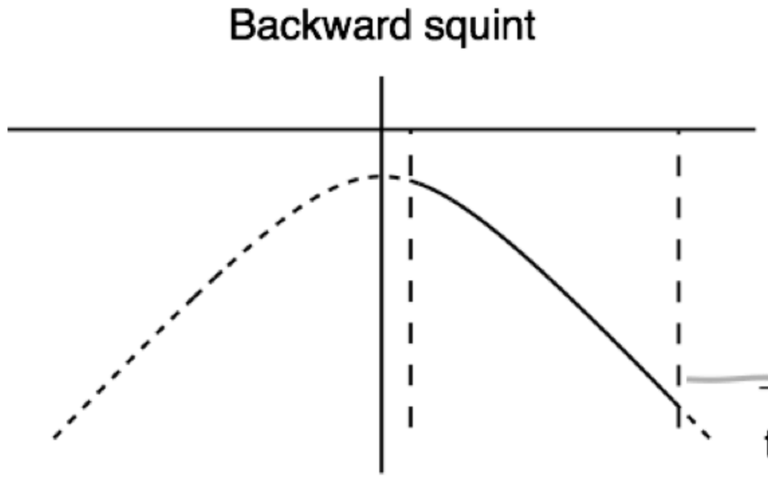
Range-Doppler Coordinates

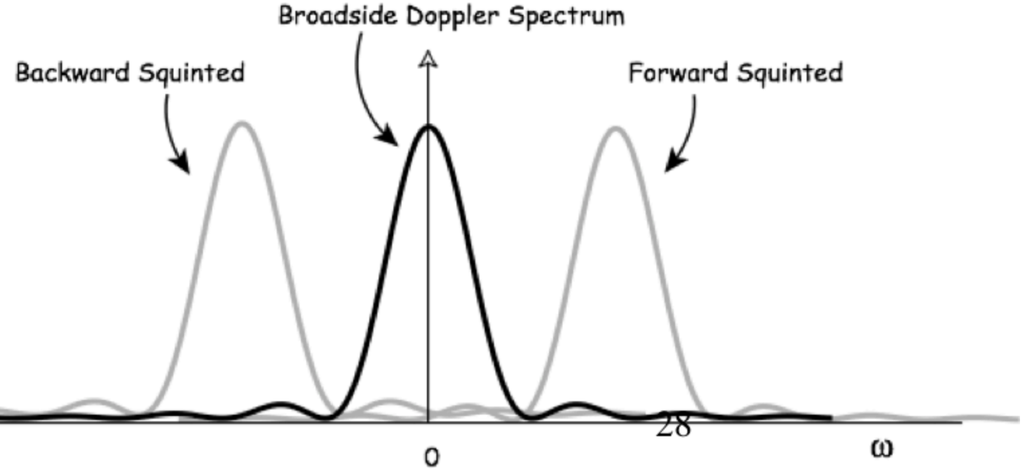


Doppler and Squint



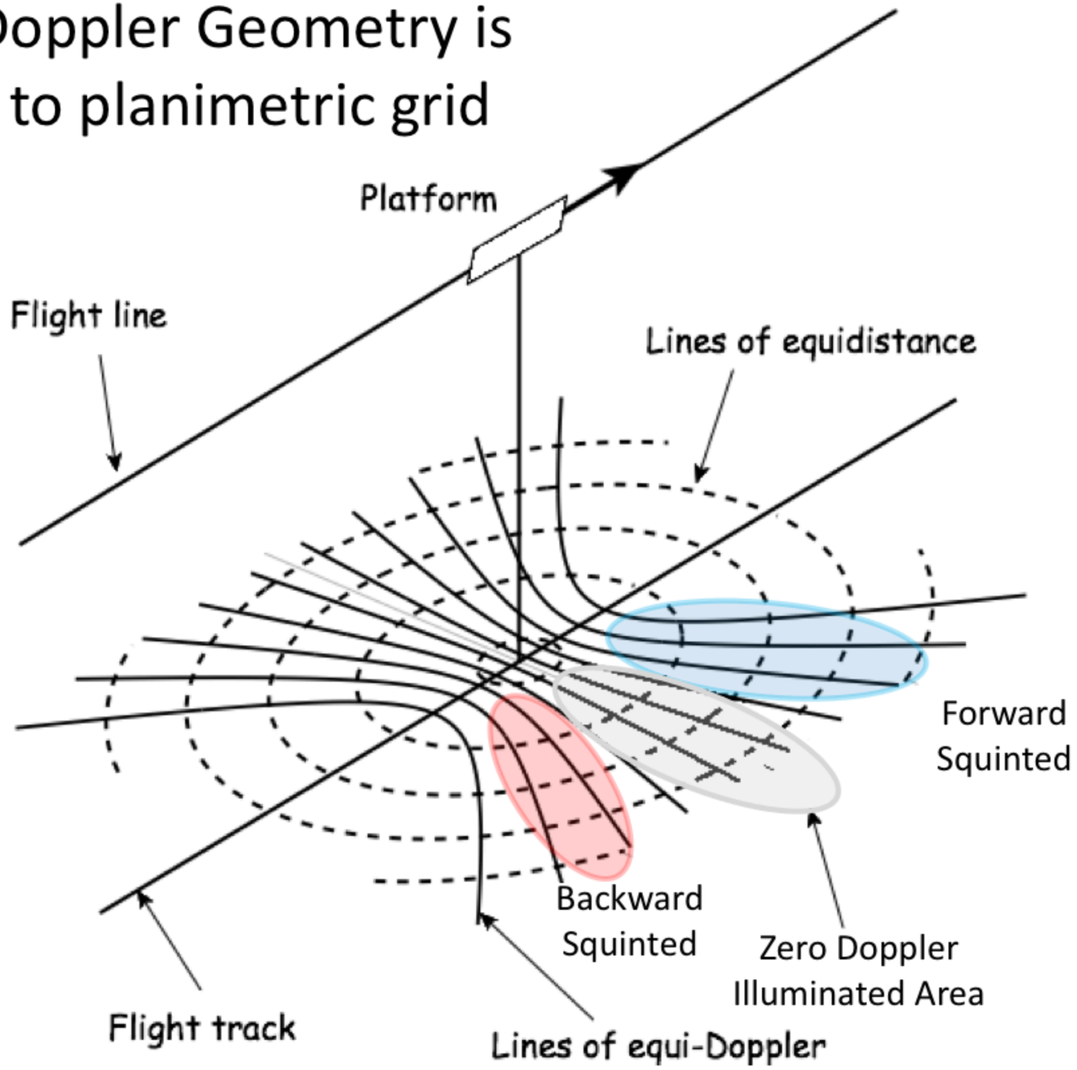






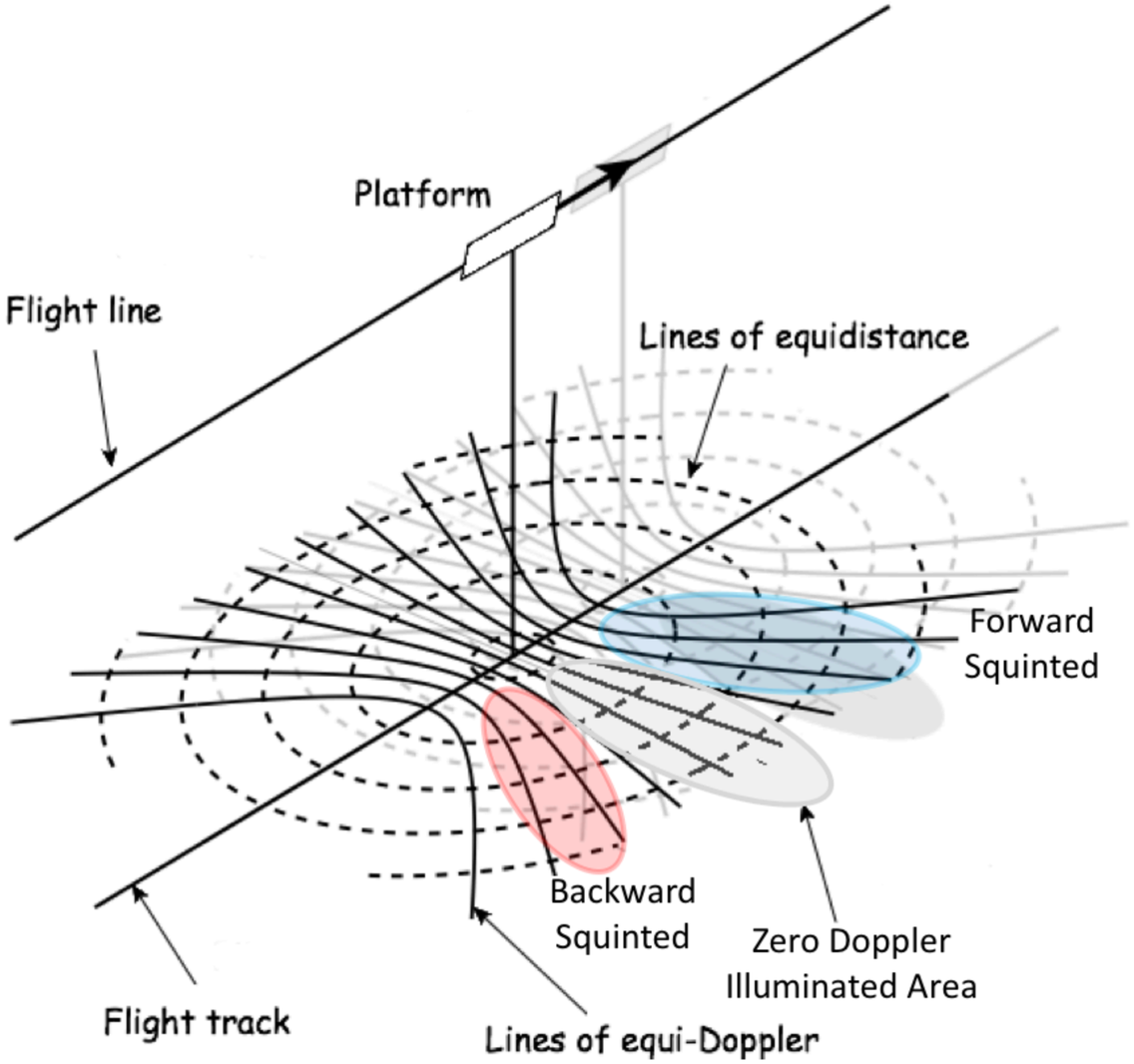
Zero Doppler Processing

Zero Doppler Geometry is closer to planimetric grid



Many prefer this output representation even if data is naturally forward or backward squinted

Zero Doppler Processing



Zero Doppler Processors output the data as though acquired from broadside. This is becoming standard.

GOING FURTHER

Learning more about SAR processing

The companion notebook **SAR Processor Tutorial** covers SAR theory and processing in much greater detail. It:

01 Simulates radar echoes from point targets

02 Develops the radar equation and the SNR of corner-reflector targets

03 Develops range and azimuth signals and processing, analytically and computationally

04 Describes linear FM signals and range compression

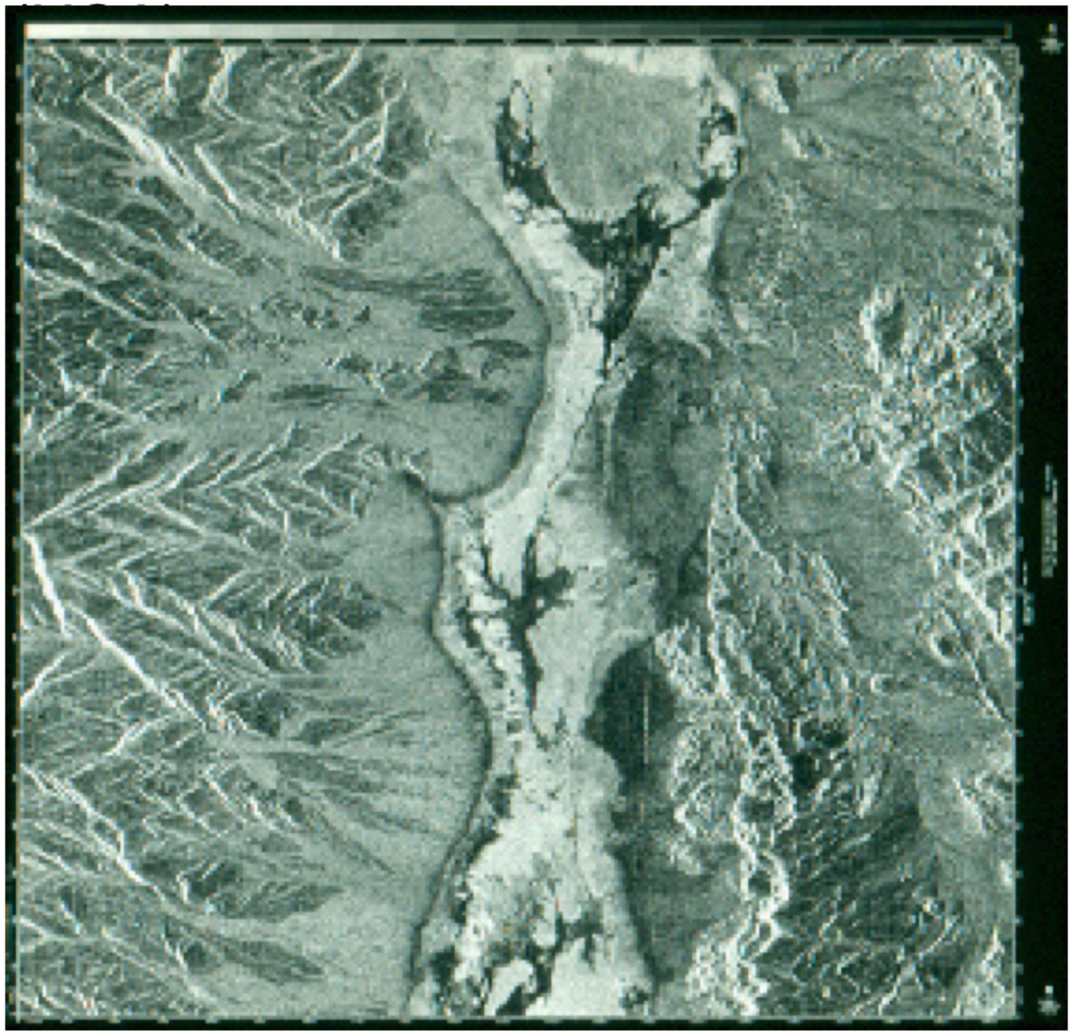
05 Describes Doppler and synthetic-aperture concepts, including range migration

06 Contrasts standard range-Doppler processing with time-domain approaches

Same course materials: 0.9_SAR_Imaging_Theory / SAR_Processor.ipynb (github.com/parosen/Geo-SInC)

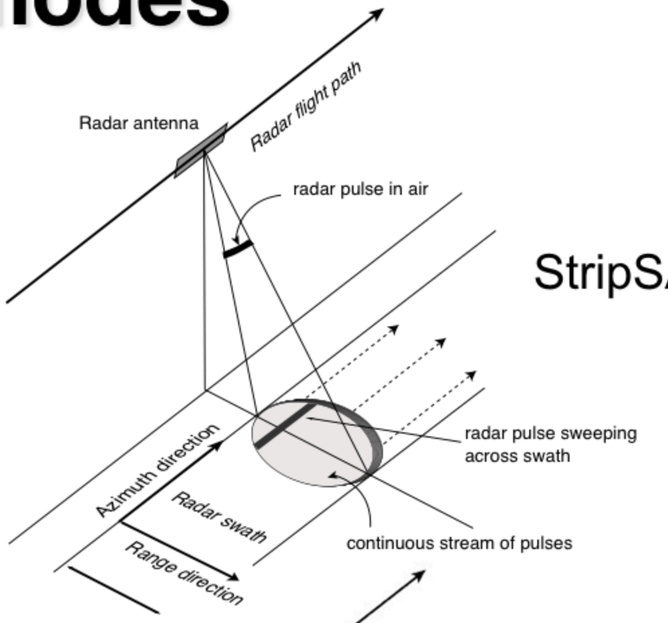
SEASAT – SAR in Space in 1978

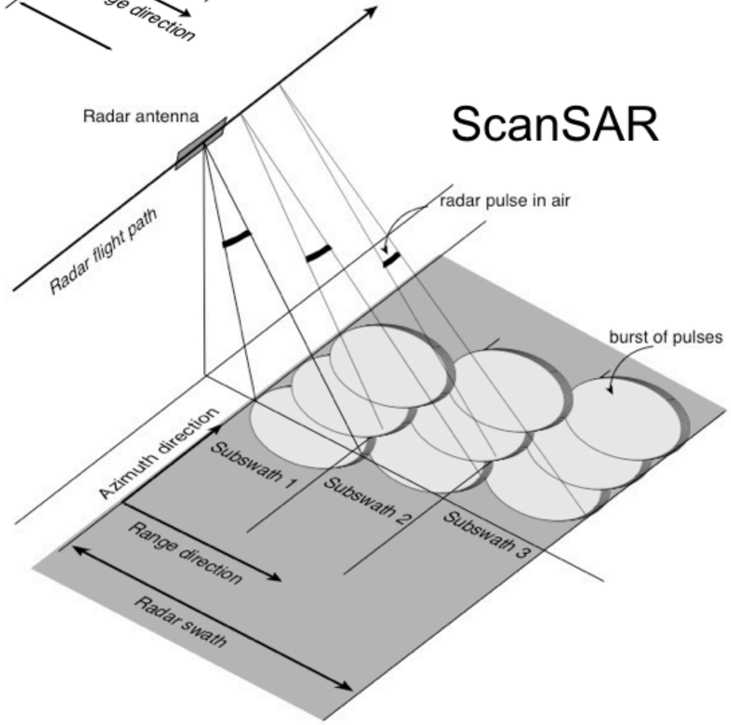
SEASAT SAR image of Death Valley



Conventional SAR modes

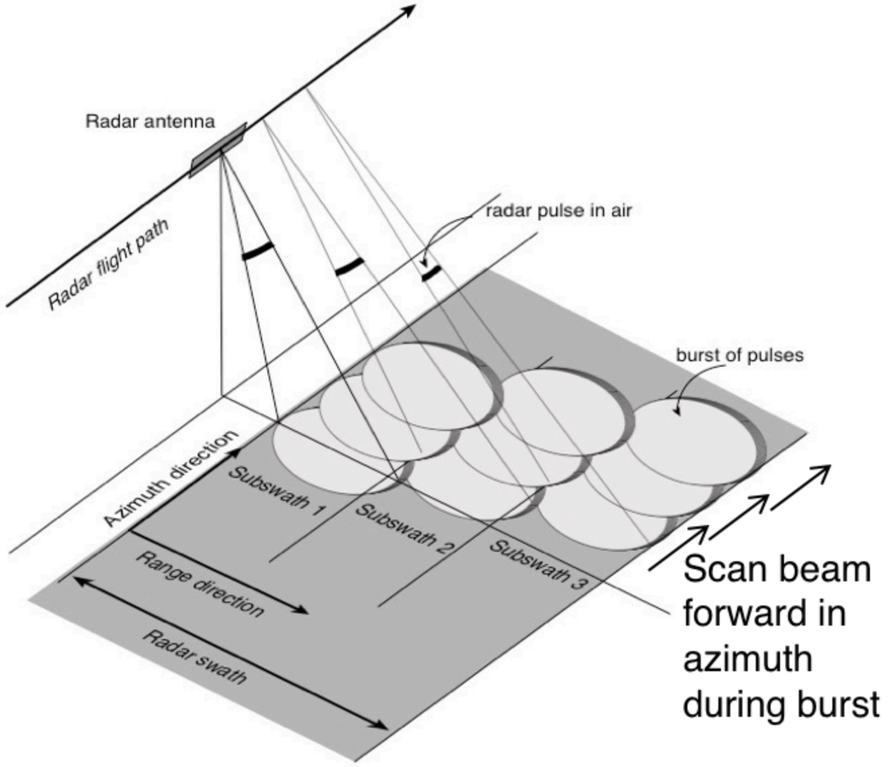
- Strip-mode SAR
  - Standard SAR mode
  - Send a pulse of energy; receive echo; repeat
  - One pulse transmit and receive at a time
  - Swath width limited by radar ambiguities
- ScanSAR
  - Wider swath low resolution SAR mode
  - Execute sequence as follows:
    - Send a pulse of energy; receive echo; repeat 50-100 times
    - Repoint the beam across-track to position 2 electrically (almost instantaneous)
    - Send a pulse of energy; receive echo; repeat 50-100 times
    - Repoint the beam across-track to position 3 electrically (almost instantaneous)
    - Send a pulse of energy; receive echo; repeat 50-100 times
  - Again, one pulse transmit and receive at a time
  - ScanSAR trades resolution (in along-track dimension) for swath: low impact on data rate
  - Generally poorer ambiguity and radiometric performance than Strip SAR





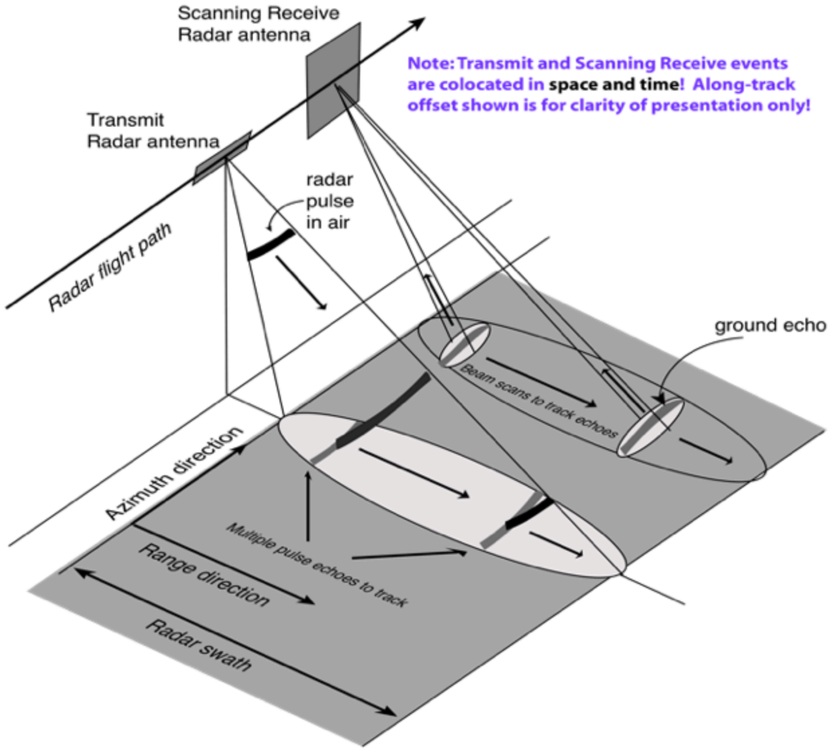
Other Wide Swath SAR Modes

Terrain Observation by Progressive Scan



- Time-share synthetic aperture among elevation beams to increase swath
- Scan beam forward in azimuth within burst to improve radiometry
- Degraded azimuth resolution

Scan on Receive SAR (SweepSAR)



- Time-share pulse returns on receive with multiple receive beams to increase swath
- Track receive echoes as they propagate across the swath
- Narrow receive beam controls ambiguities

02

PART 2 · RADAR IMAGING FUNDAMENTALS

Properties of radar images

What radar measures · scattering · imaging geometry · wavelength · polarization · radar equation · speckle

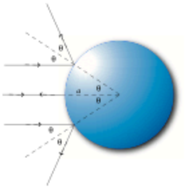
What Radar Can Tell Us

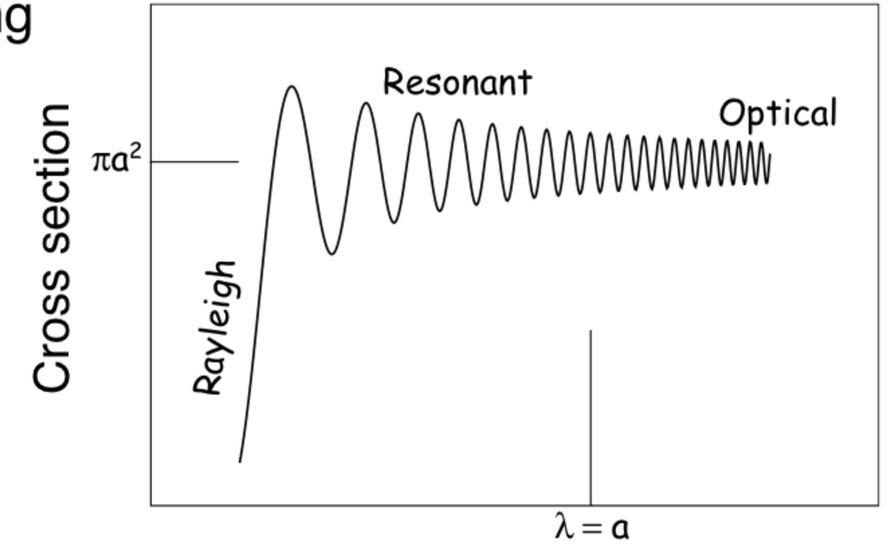
- Transmitted radar signals have known characteristics
  - Amplitude
  - Polarization
  - Phase and Time Reference
  - Wavelength, or Frequency
- A distant object that scatters the radar signal back toward the receiver alters the amplitude, polarization, and phase, differently for different wavelengths
- Comparison of the received signal characteristics to the transmitted signal allows us to understand the properties of the object.

Surface and Volume Scattering Models for Radar

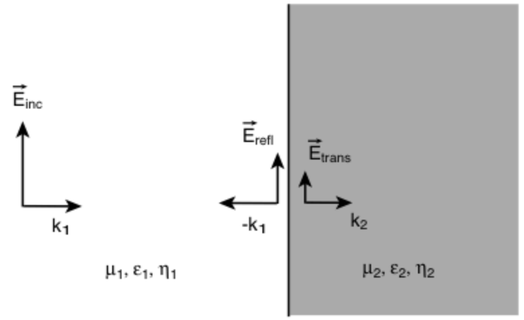
We can model simple scattering from particles or surfaces…

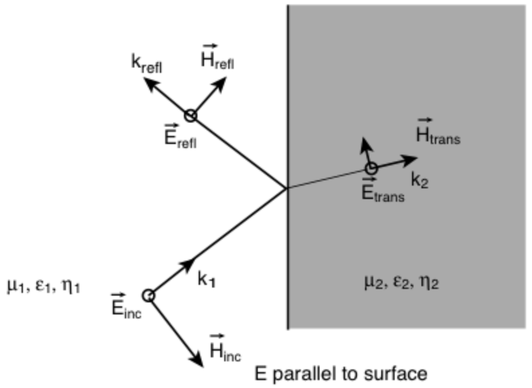
Cross section of a large sphere is its projected area



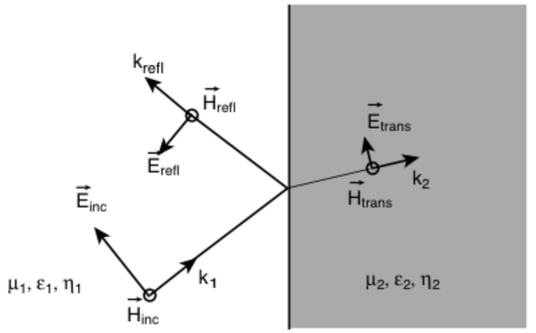


Cross section of a large flat facet goes as area squared



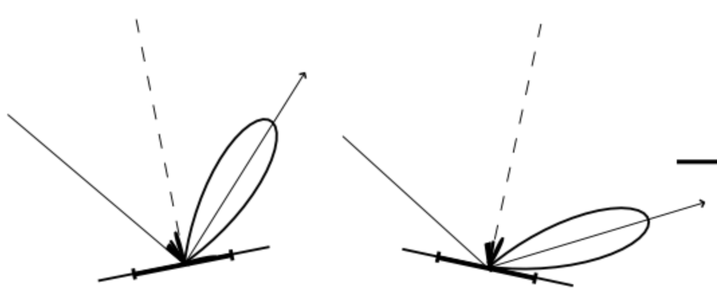


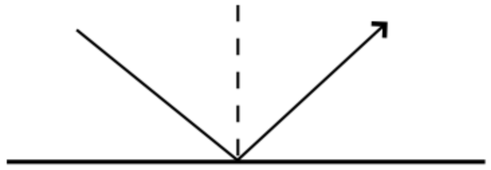
E parallel to surface

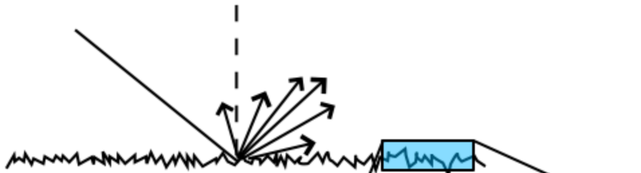


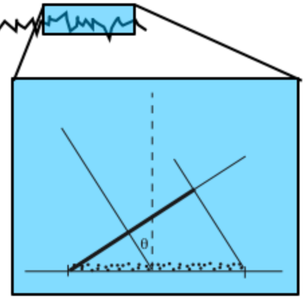
H parallel to surface

Surface and Volume Scattering Models for Radar

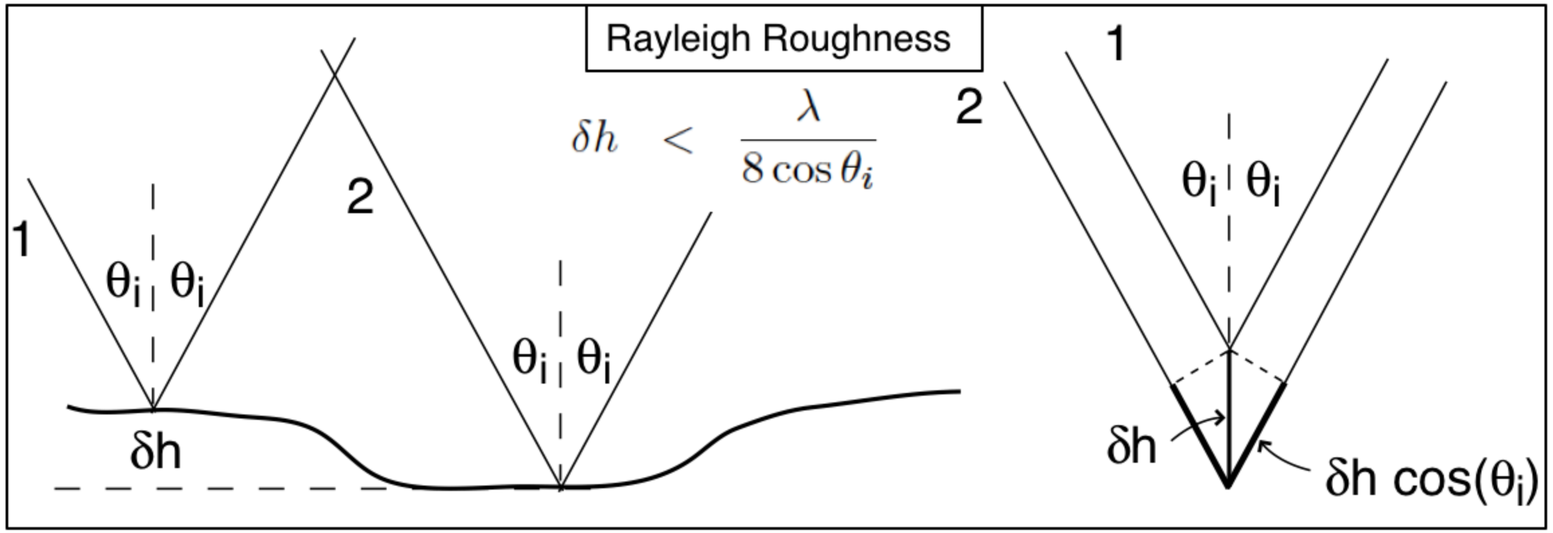




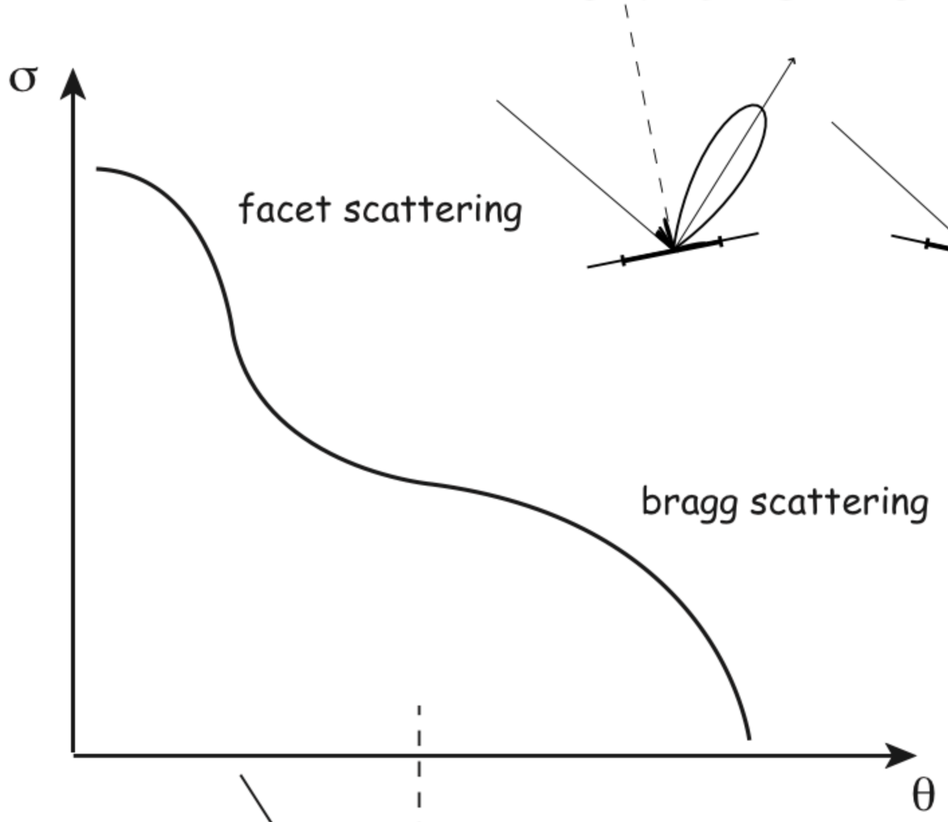




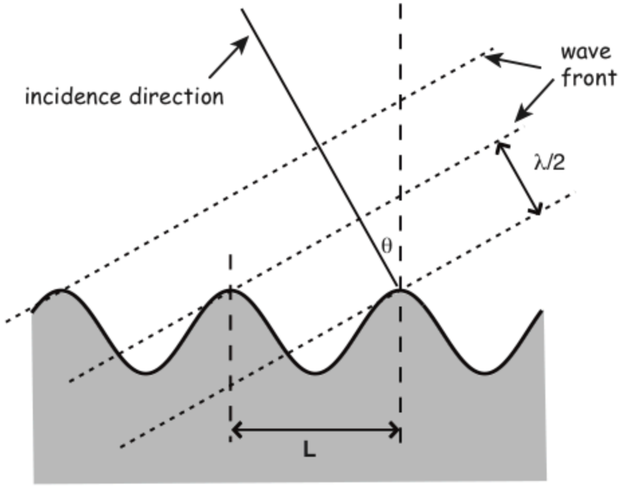
And we can develop intuition from combinations of these elements…

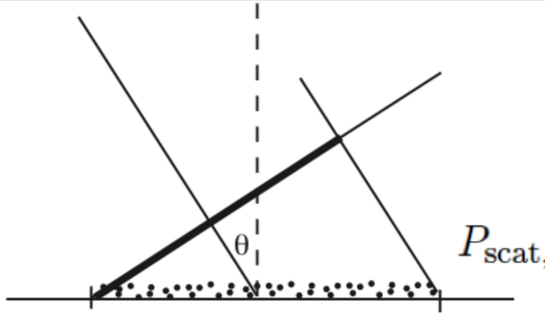


Surface and Volume Scattering Models for Radar



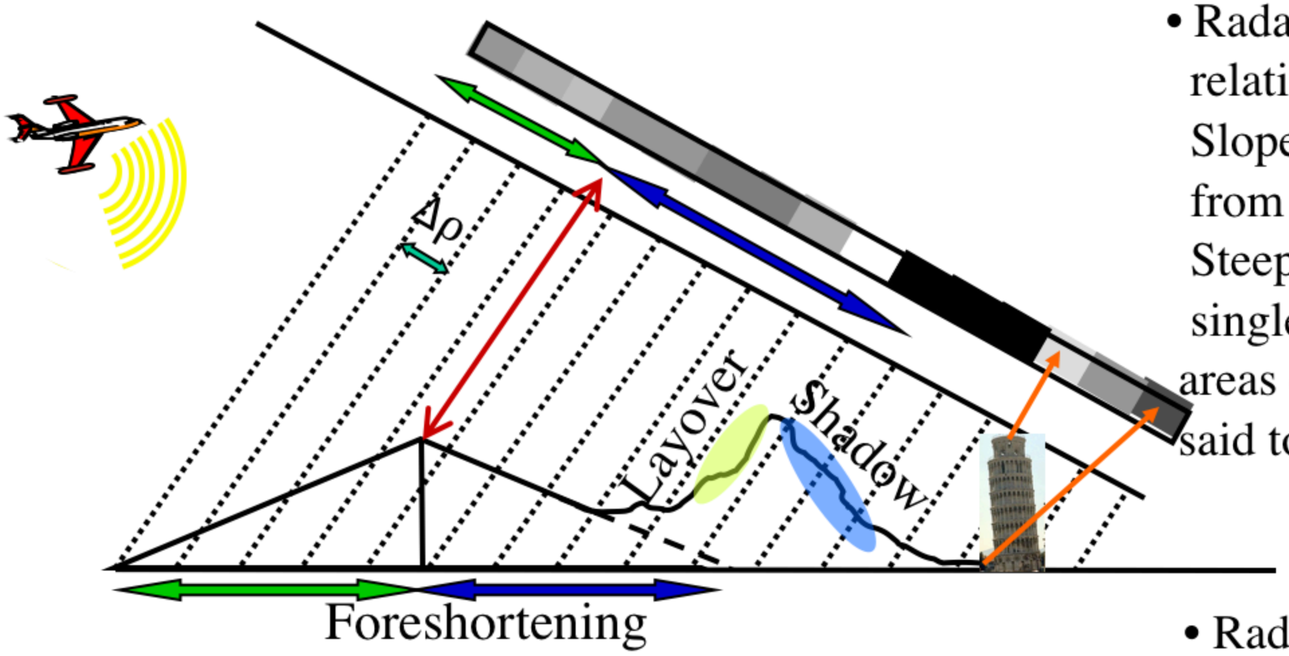
$$\sigma_{\text{surface}}(\theta) = \Gamma^2_{\text{Refl}} \sum_j p^2(\theta, \theta_{\text{inc}} - \alpha_j)$$



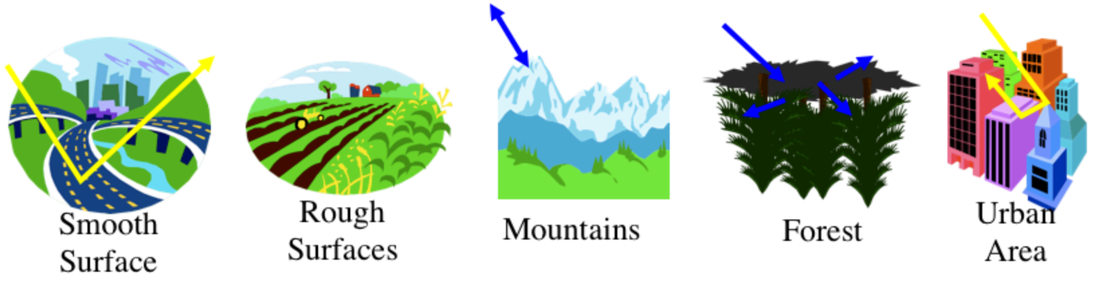


$$P_{\text{scat,d}} = \frac{P_{\text{inc,d}}}{4\pi r^2} \sigma_{\text{point}} \cos\theta_i N$$

Radar Imaging Properties



- Radar images are distorted relative to a planimetric view. Slopes facing toward or away from the radar appear foreshortened. Steep slopes are collapsed into a single range cell called layover and areas occulted by other areas are said to be shadowed.



- Radar is primarily sensitive to the structure of objects being imaged whereas optical images are primarily sensitive to chemistry.
- The scale of objects relative to the radar wavelength determines how smooth an object appears to the radar and how bright or dark it is in the imagery.

Wavelength - A Measure of Surface Scale

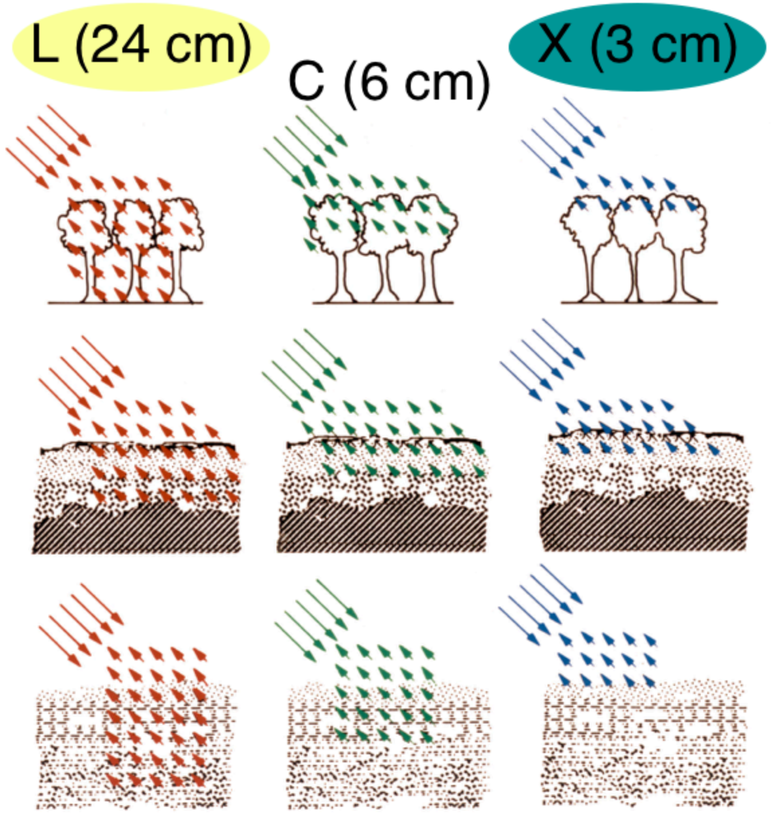
Light interacts most strongly with objects on the size of the wavelength

L (24 cm) C (6 cm) X (3 cm)

Forest: Leaves reflect X-band wavelengths but not L-band

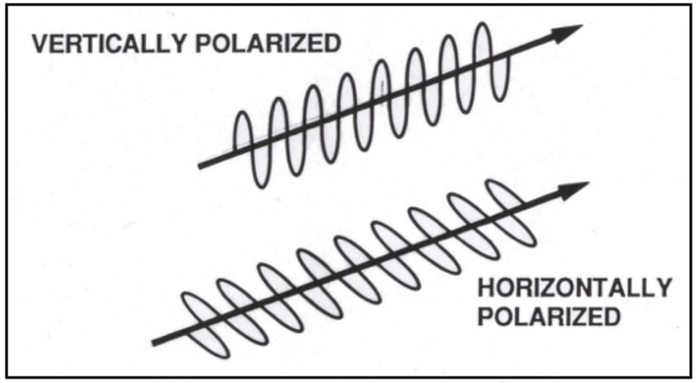
Dry soils: Surface looks rough to X-band but not L-band

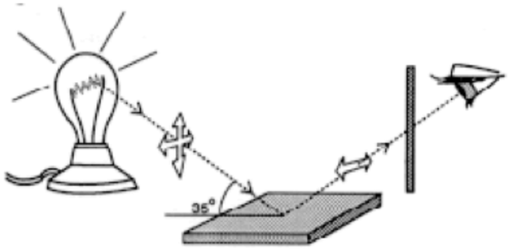
Ice: Surface and layering look rough to X-band but not L-band



Polarization - A Measure of Surface Orientations and Properties

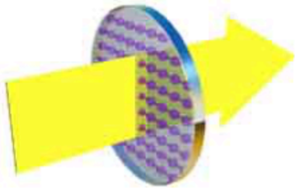
Wave Polarization



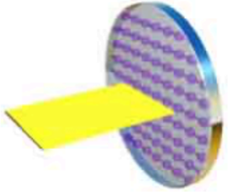


Mostly horizontal polarization is reflected from a flat surface.

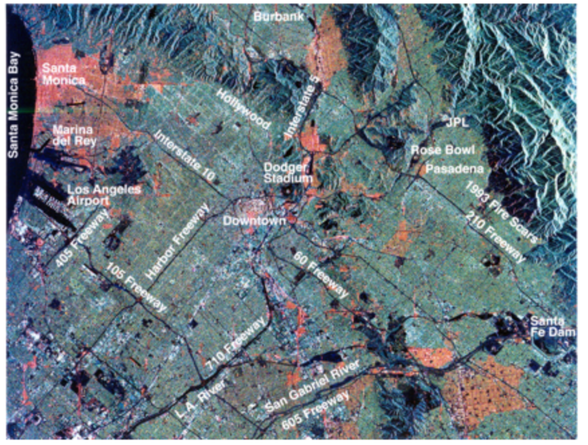
Polarization Filters

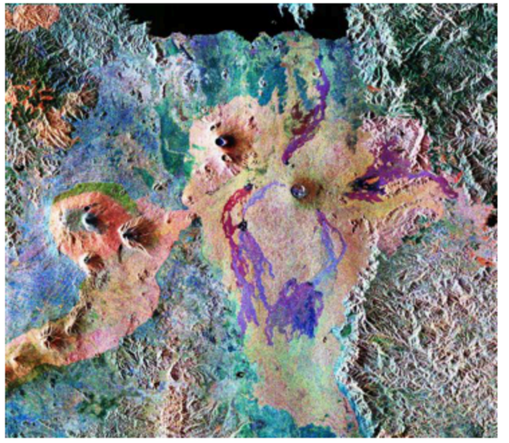


Vertical polarization passes through horizontally arranged absorbers.



Horizontal polarization does not pass through horizontally arranged absorbers.





Color figures from www.colorado.edu/physics/2000

Radar Equation in Remote Sensing

- Backscatter can be used to interpret surface properties
  - Often using wavelength and polarization to probe characteristics
- Other radar instrument characteristics affect the science performance
  - Wavelength and antenna size determine illuminated area and real-aperture resolution
  - Antenna size in flight direction determines resolution for a synthetic aperture radar
  - Antenna area determines ambiguity, or aliasing, performance and swath extent
- Sensitivity of the measurement determined by “radar equation”

$$\mathrm{SNR} = P_T \cdot G_T(\lambda) \cdot \frac{1}{4\pi R^2} \cdot \sigma(\lambda) \cdot \frac{1}{4\pi R^2} \cdot A_R \cdot \epsilon(\lambda) \cdot \frac{1}{kTB}$$

Signal-to-Noise Ratio

Transmit Power

Transmit Antenna Gain

Range (distance)

Radar cross-section or “Reflectivity”

Receive Antenna Area

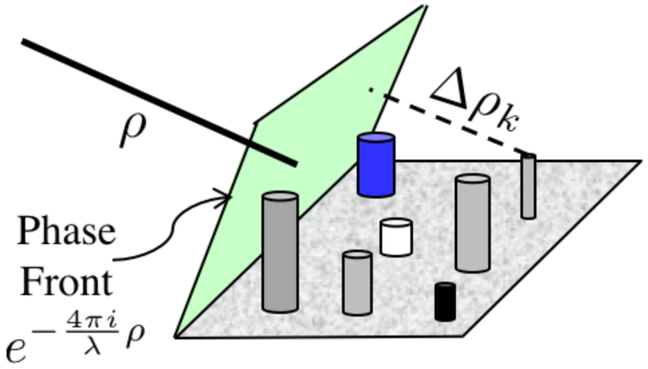
System Efficiency/Losses

Receiver Temperature

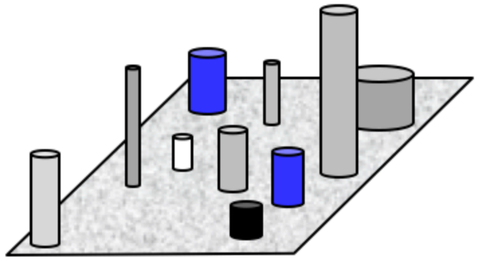
Receiver Bandwidth

SAR Imagery and Speckle

- Full resolution SAR imagery has a grainy appearance called speckle, which is a phenomena due to the coherent nature of SAR imaging.
- The total signal return from a resolution cell is the coherent sum of the returns from all the individual scatterers within it that are generally randomly distributed.



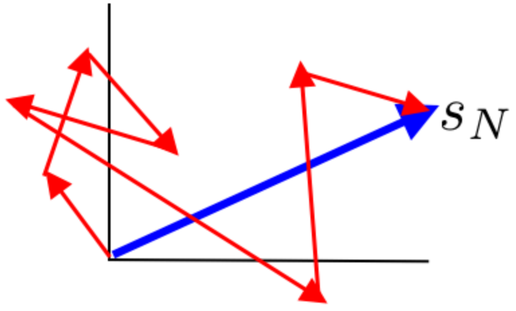
Number and arrangement of scattering elements within resolution cell varies from pixel to pixel.



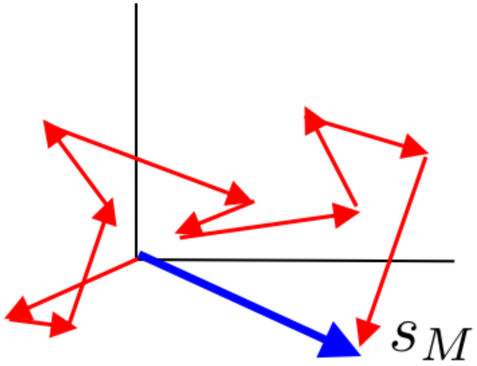
Returned signal is a coherent combination of the returns from the scattering elements.

Pixel N

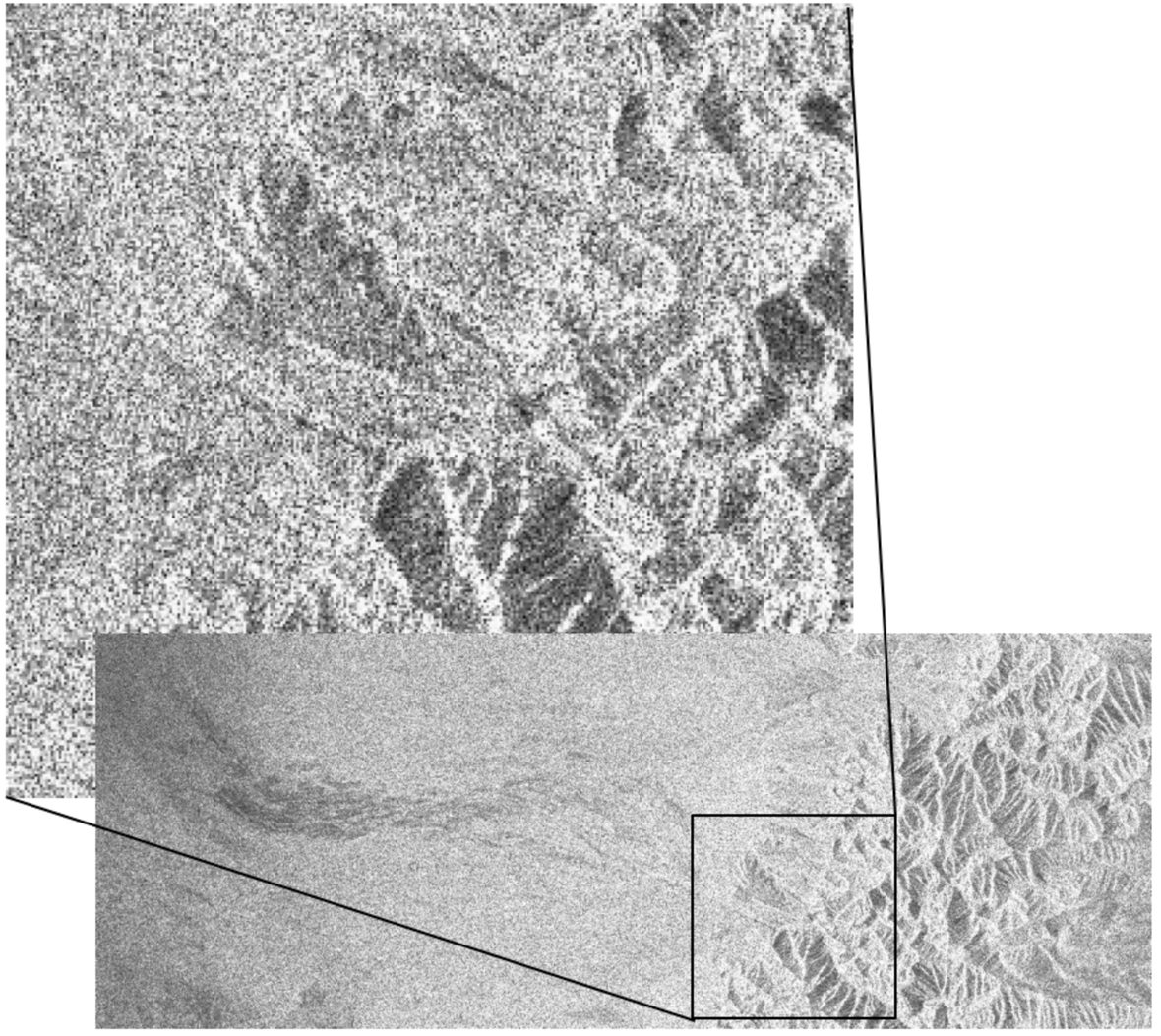
Pixel M



$$s = A \underbrace{e^{-\frac{4\pi i}{\lambda}\rho}}_{\text{Range Phase}} \underbrace{\sum_{k=1}^{N} a_k e^{-\frac{4\pi i}{\lambda}\Delta\rho_k}}_{\text{Scatterer Contribution}}$$



Speckle and Radiometric Accuracy



Natural “laser” speckle masks intrinsic radar backscatter which contains surface information, independent of SNR!

Spatial averaging (looks) reduces the speckle and draws out the natural backscatter reflectivity

COURSE MATERIALS ON GITHUB

Geo-SInC: the InSAR short-course repository

Repository

github.com/parosen/Geo-SInC

Free Jupyter notebooks from the EarthScope (formerly UNAVCO) InSAR Processing and Analysis short course, 2021–2024. Today's lecture is **0.8_SAR_Theory_Phenomenology / SAR.ipynb**.

PRE-COURSE

Python, GIS with GDAL, data access and SAR theory (this lecture)

DAY 1

Geophysical modelling, InSAR theory, stripmap processing with ISCE2

DAY 2

Interpreting coseismic interferograms; TOPS (Sentinel-1) processing

DAY 3

Glaciers, preparing data for modelling, atmospheric errors

DAY 4

Time-series theory, stacks with ARIA-tools, offset stacks

DAY 5

Time-series analysis with MintPy and stack processing with ISCE2

Folders by year: UNAVCO2021, UNAVCO2022, EarthScope2023 and EarthScope2024 (newest). To get everything, figures included: on GitHub, choose Code → Download ZIP.

Thank you. Questions?

Content adapted from Paul A. Rosen, “SAR Principles and Phenomenology”, InSAR short-course notebook (2021), github.com/parosen/Geo-SInC.
Contributors: Scott Hensley, Anthony Freeman, Jakob van Zyl, Piyush Agram, Howard Zebker. Animation: NISAR mission.